



“ Make creating your own songs
within reach.”

- **Area:**
Music Products, UI Design, HCI
- **Feature:**
Experiment, entertaining
- **Duration:**
Jan.2024 - May.2024
- **Solo project**



Facts & Ethos

As a music enthusiast with ten years of piano experience, I have a strong desire to further develop my skills in songwriting. However, despite having a solid foundation in music theory, I still find it difficult to begin composing music. The knowledge required for arrangement is complex and tedious.

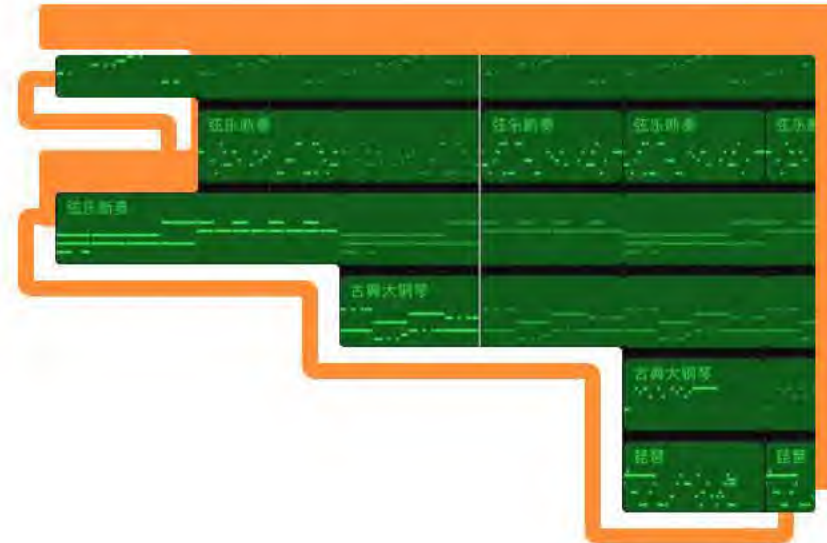


Rhythm Box

Simply drag various symbols representing **different sounds** onto the character, and it will blend into a wide range of music with strong rhythmic sensations.

Children's Music Theory Teaching Toy

Using fun methods such as pizza pie puzzles, the children's music theory teaching is flexible and engaging, allowing for various practice combinations. However, all tools are paper-based, with rhythm and tempo sung independently, and there's a lack of auditory teaching aids.



Garage Band

It allows for quick composition of songs with a wide selection of accompaniments on different tracks. Popularizing the concept of tracks, but cannot be used as an introductory software for music arrangement; the operable areas in music are too limited.

Note Cards

Due to the extensive and scattered nature of music theory knowledge itself, the accessories for these teaching tools are small and numerous, making storage and operation inconvenient.

However,
after personally using them, I found that they fail to genuinely support the creation of songs

Lacks Long-term Appeal

Insufficiently Visualized

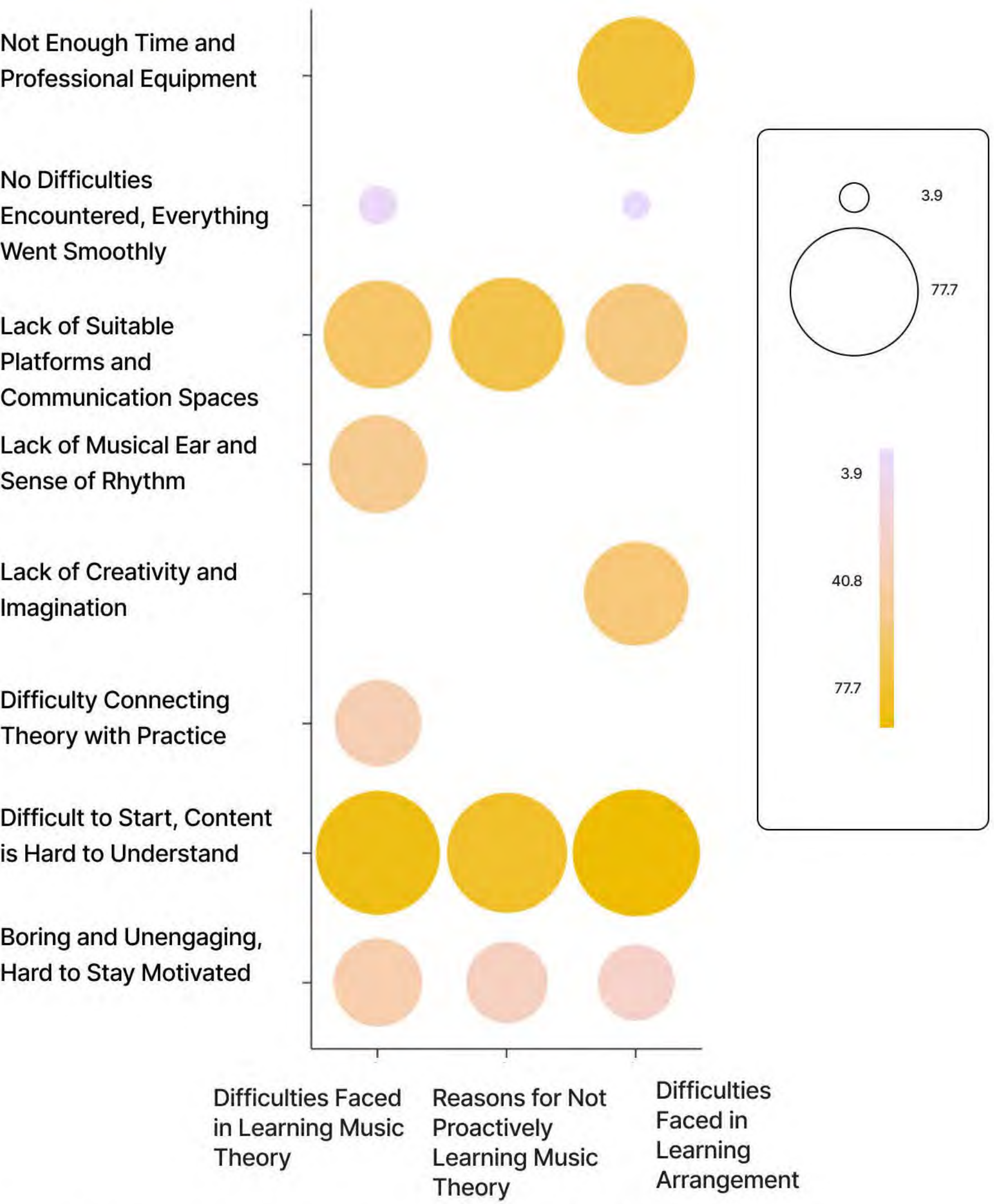
Unable to Truly Achieve Free Composition in Arrangement

Stuck at "Fusion" rather than "Creation"

RESEARCH

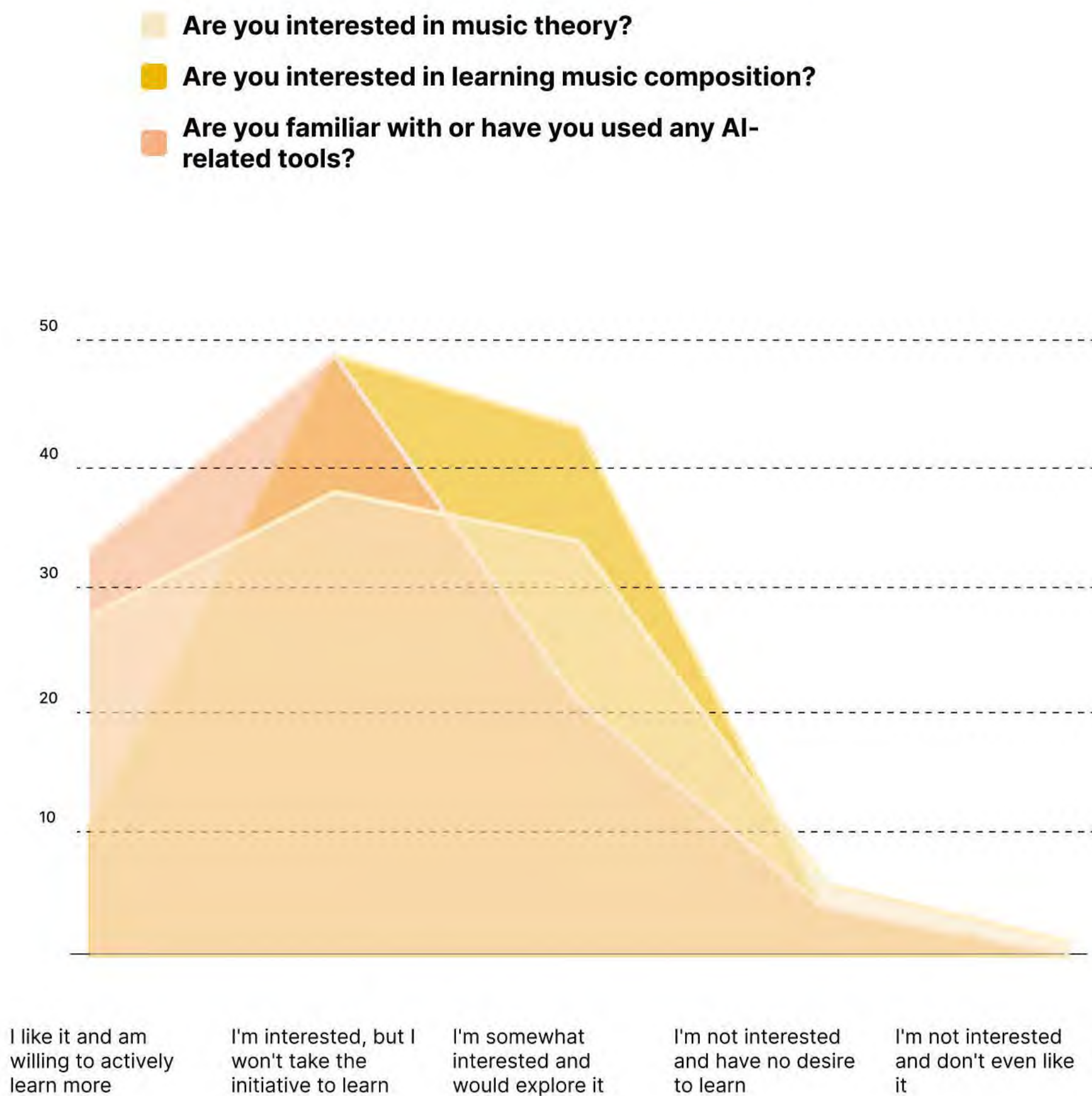
THIS QUESTIONNAIRE CONSISTS OF 19 QUESTIONS, WITH 107 PARTICIPANTS, AND 106 VALID RESPONSES.

1 QUESTIONARE 1 DIFFICULTIES AND SEVERITY IN MUSIC THEORY



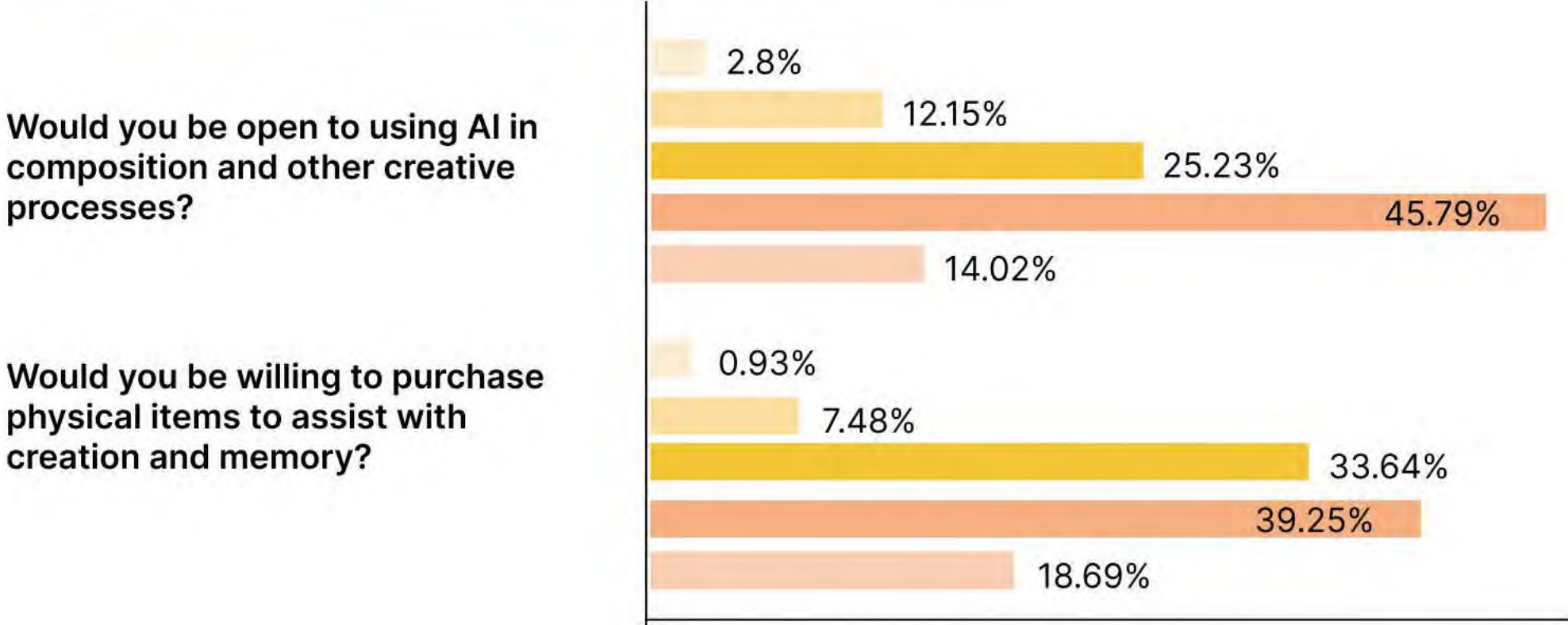
The main challenges in learning **music theory** include difficulty in understanding the content, lack of suitable platforms and communication spaces, boredom, lack of musical and rhythmic sense, and difficulty in integrating theory with practice.

2 QUESTIONARE 2 LEVEL OF INTEREST AND FAMILIARITY



Regarding music theory and related knowledge, many people have limited understanding, often only hearing about it without actual exposure. However, the majority show a positive attitude towards learning music theory and composition, with only 5% expressing disinterest or a lack of desire to learn.

3 QUESTIONARE 3 ATTITUDE TO MY PROPOSAL

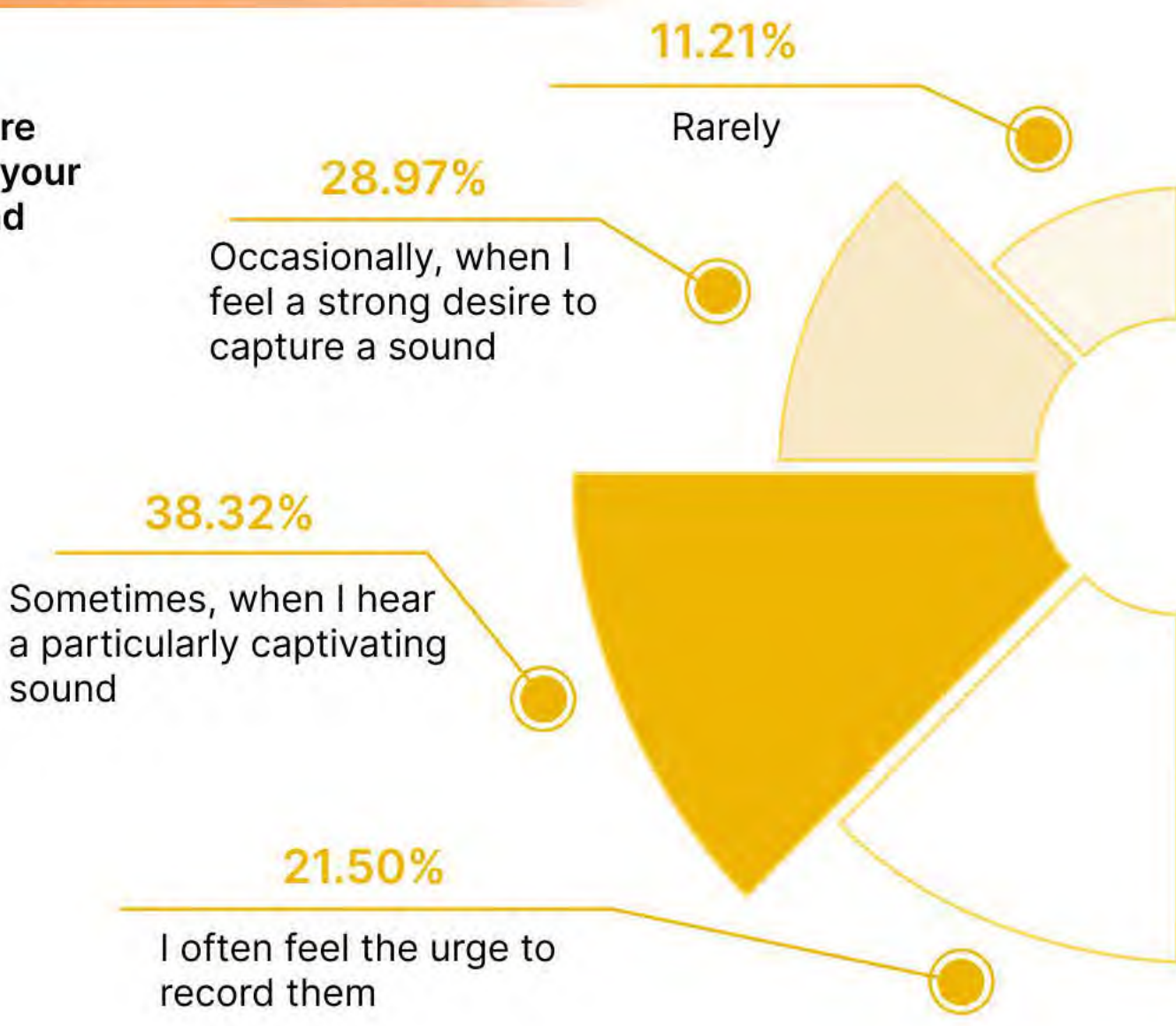


Explored multiple aspects of song creation. Users showed high acceptance of the combination of physical items and online platforms, as well as AI-assisted composition, which provided direction for future design.

4 QUESTIONARE 4 EVERYDAY SOURCES OF INSPIRATION

Have you ever wanted to capture certain rhythms or melodies in your daily life (e.g., the sound of wind chimes, or even a car horn)?

Most people suggested recording environmental sounds to inspire composition, incorporating them into the song as part of the creative process.



INTERVIEW



Interview Participants:
12 people, Ages 8-55,
Occupations including students,
Employees, Retired Workers

Music Theory Foundation:

- No prior exposure to music theory
- Knowledge of basic music theory
- Learned multiple instruments, attempted self-arranging

Would diverse visual representations help you continue composing?

I would like to have personalized visuals that are unique to my songs. This would enhance my interest in using the platform for composition.

If there were interactive physical items for creation, what kind of experience would you hope to have?

It should have a memory function to capture my inspirations at any time, and it should have a sense of rhythm when used.

What would spark your interest in composing music?

I think there need to be some irresistible features that keep me engaged, so I can continue playing.

When I struggle to connect parts of a composition, it should provide hints and assistance, offering visuals or melodies to inspire new ideas.

If you try composing music, what challenges would you face, and what form would you prefer to try?

What would motivate you to keep creating music?

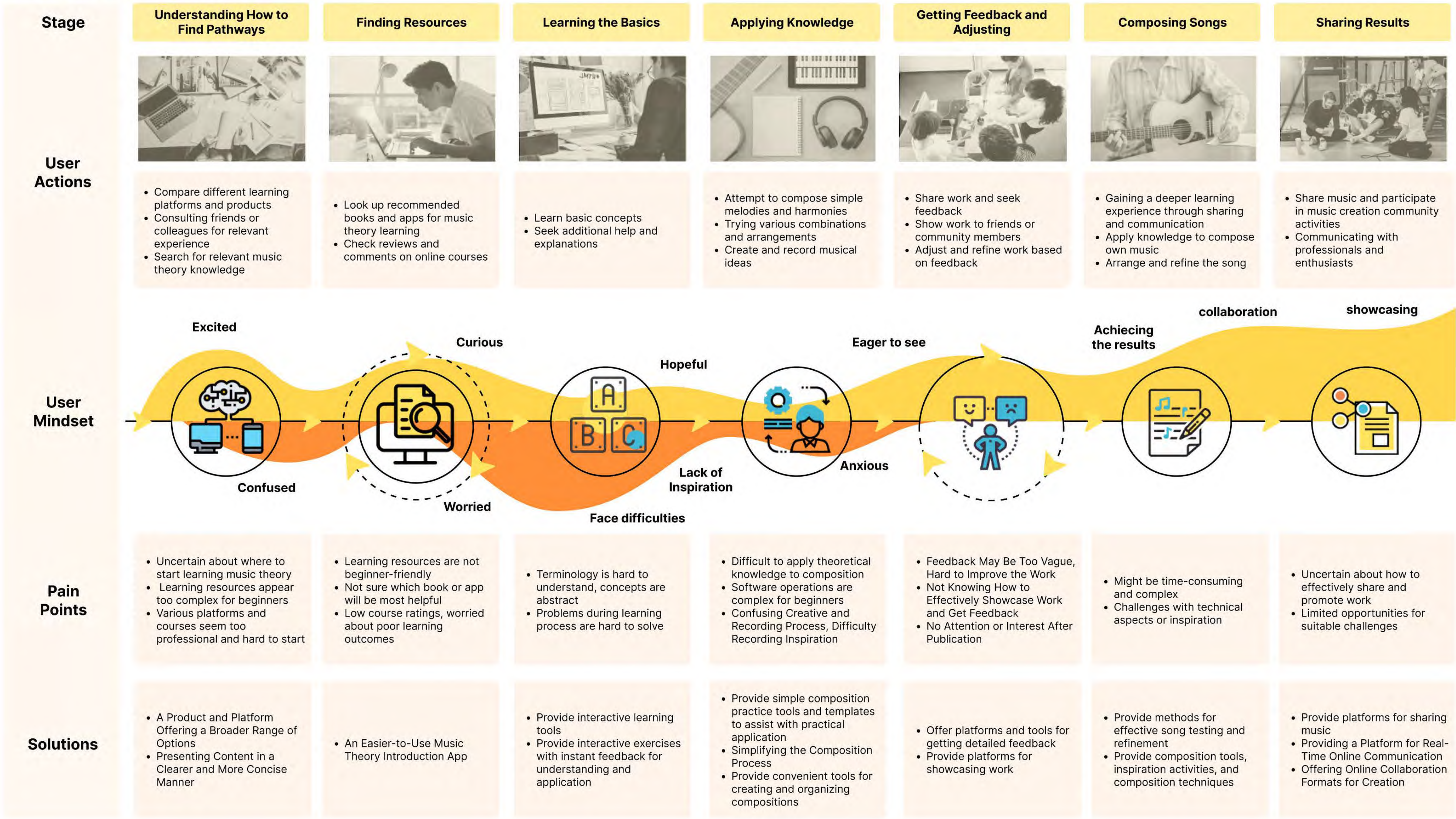
I enjoy finding inspiration in music salons, where I can appreciate music with others, exchange ideas, and tell stories, which sparks creativity for songwriting.

I also want to share the story behind the songs I create, the emotions I felt while composing, along with corresponding visuals.

There definitely needs to be something visual, something that allows me to clearly see the changes in the music, so I can learn and create on my own.

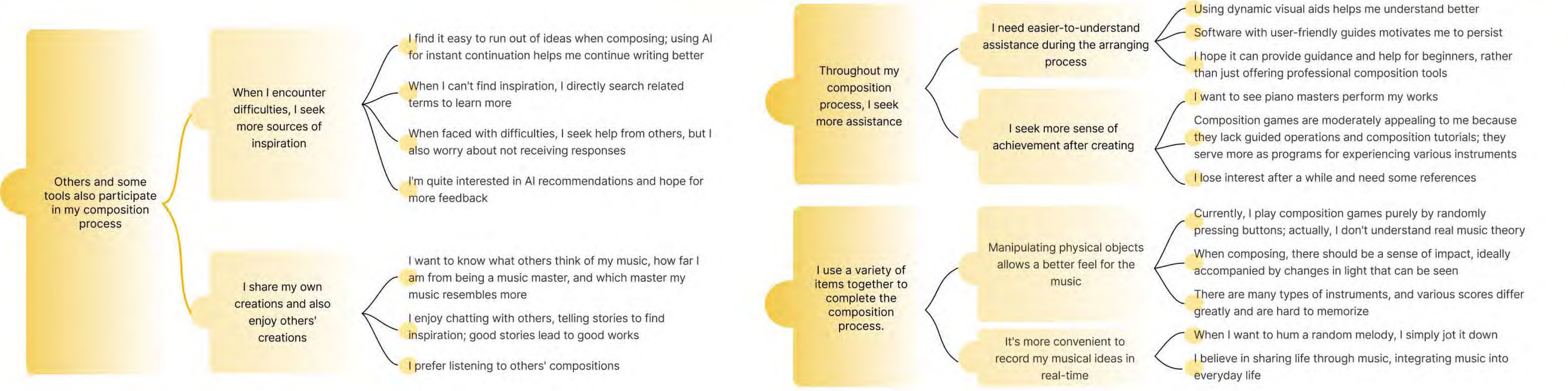
Insight : Conducted in-depth interviews on interest in songwriting, challenges faced, and desired experiences. Everyone agreed that learning to create songs is not easy and requires significant time investment. During the learning process, having visualized space and dedicated creation platforms is essential. Additionally, getting stuck in the composition process was identified as a key reason for halting progress.

JOURNEY MAP



AFFINITY DIAGRAM

NEEDS



PAINTPOINTS

Name: Wang Xuehua
Age: 25
Occupation: Graduate student
Background: Has learned musical instruments and is eager to further explore composition



Features

- Has a basic understanding of music theory and knowledge
- Enjoys rhythm games and percussion musical toys in her free time
- Very interested in creating her own music compositions

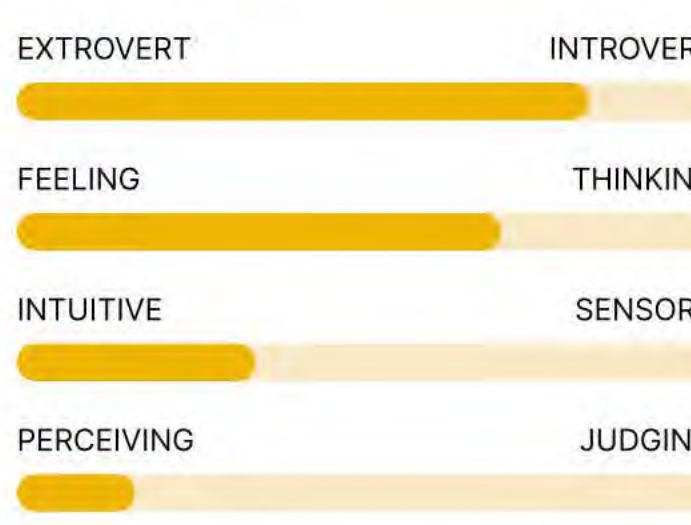
Pain Points

- Often unable to progress further due to a lack of inspiration sources
- Many melodies and ideas are hard to record conveniently
- Existing platforms have additional barriers and are hard to get started with

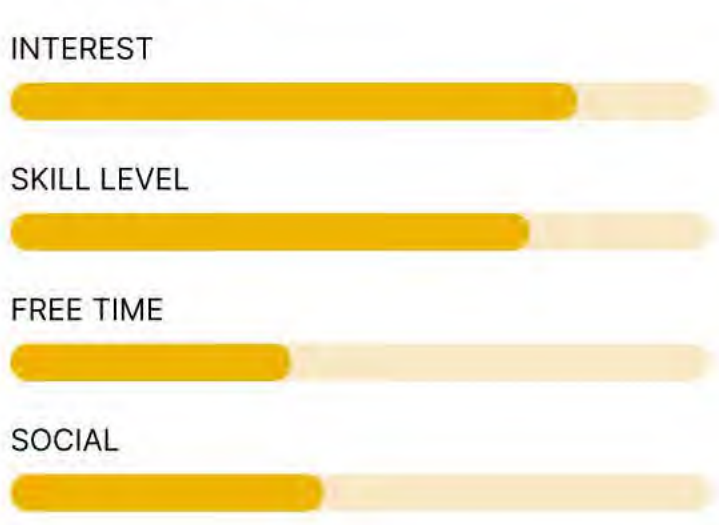
Needs

- A platform for composition and saving work
- More creative and autonomous tools for arranging music
- A way to record everyday sounds and incorporate them into compositions

PERSONALITY



MOTIVATION



Name: Qin Lan
Age: 58
Occupation: Retired worker
Background: Has some basic knowledge of musical instruments but lacks a deep understanding of music theory



Features

- About to retire, willing to develop new hobbies
- Enjoys listening to music while drinking tea and wishes to compose his own songs
- Has a lot of free time

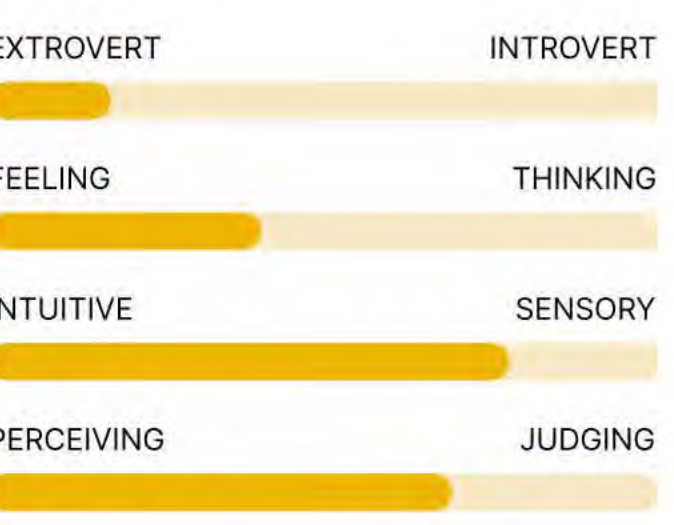
Pain Points

- Difficulty understanding music terminology and concepts
- Complex online platforms and usability issues
- Lack of communication channels for interaction

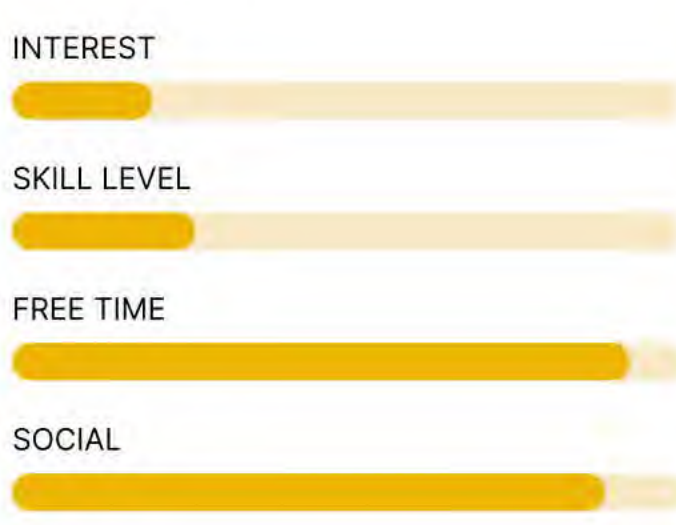
Needs

- Quick and easy understanding of music theory and composition
- Emphasis on the storytelling aspect of music
- Smart tools for refining compositions and expressing melodies

PERSONALITY



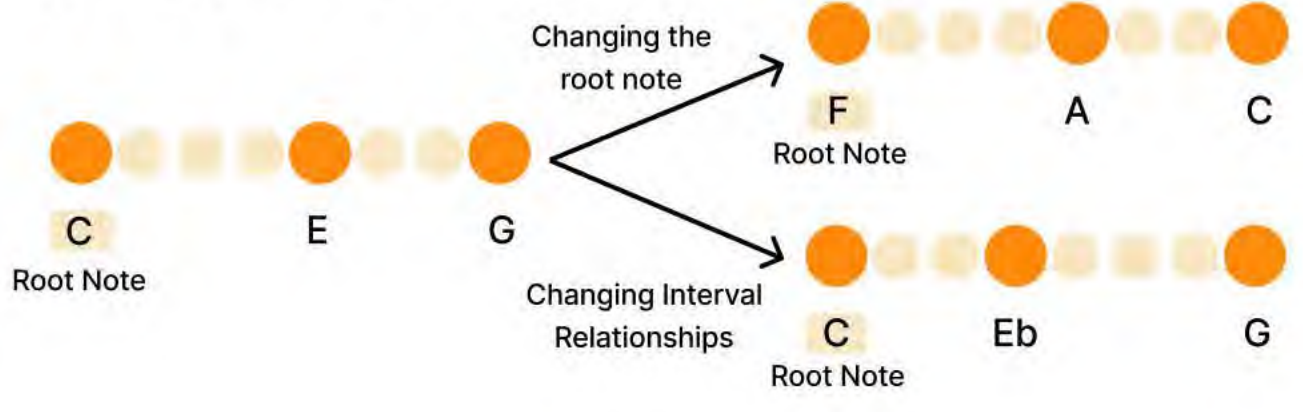
MOTIVATION



DESIGN PROCESS

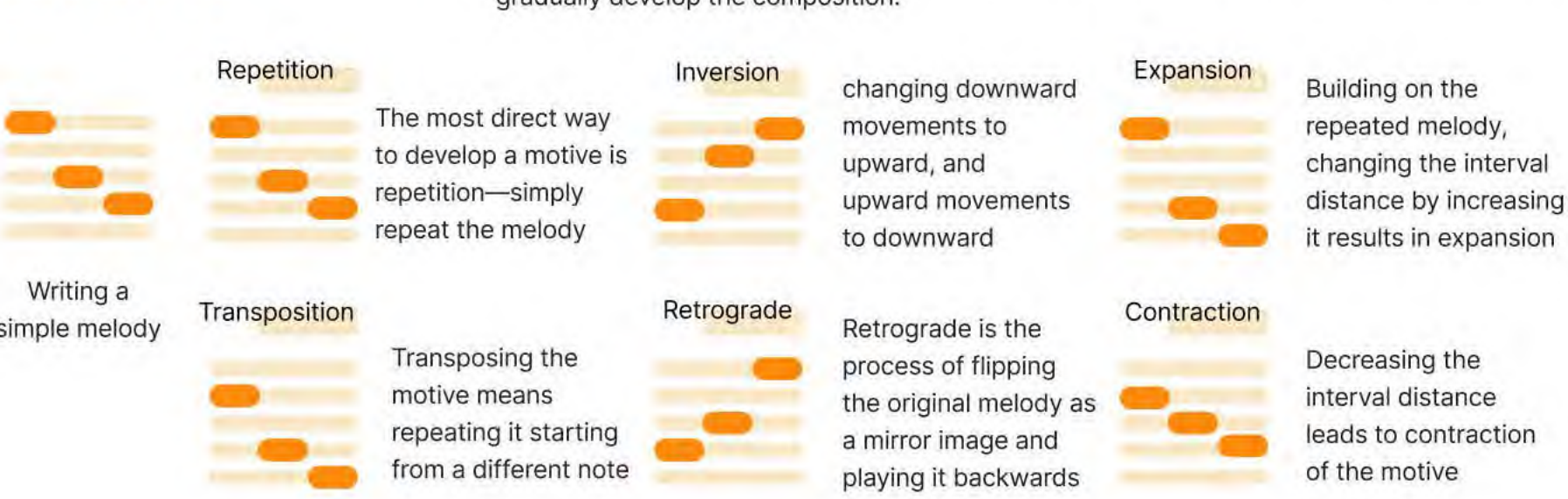
Chords	C	D	E	F	G
Major	C (C-E-G)	D (D-F#-A)	E (E-G#-B)	F (F-A-C)	G (G-B-D)
Minor	Cm (C-Eb-G)	Dm (D-F-A)	Em (E-G-B)	Fm (F-Ab-C)	Gm (G-Bb-D)
Dominant 7th	C7 (C-E-G-Bb)	D7 (D-F#-A-C)	E7 (E-G#-B-D)	F7 (F-A-C-Eb)	G7 (G-B-D-F)
Diminished 7th	Cdim7 (C-Eb-Gb-Bbb)	Ddim7 (D-F-Ab-Bbb)	Edim7 (E-G-Bb-Db)	Fdim7 (F-Ab-Cb-Ebb)	Gdim7 (G-Bb-Db-Fb)
Half-diminished 7th	Cm7(b5) (C-Eb-Gb-Bb)	Dm7(b5) (D-F-Ab-C)	Em7(b5) (E-G-Bb-D)	Fm7(b5) (F-Ab-C-Eb)	Gm7(b5) (G-Bb-Db-F)

There are several common chord combinations that are easy to use and sound harmonious. However, there is a wide variety of chords, and many great songs use dissonant chords to create striking effects through unique expressions. Experimenting with different combinations might help you enjoy music more and experience something unique.



Chords may seem complex, but they are primarily composed of two parts: the root note and the chord type. Once the root note is determined, changing the chord type (interval relationship) results in a new chord. Similarly, by fixing the interval and changing the root note, a new chord can also be created.

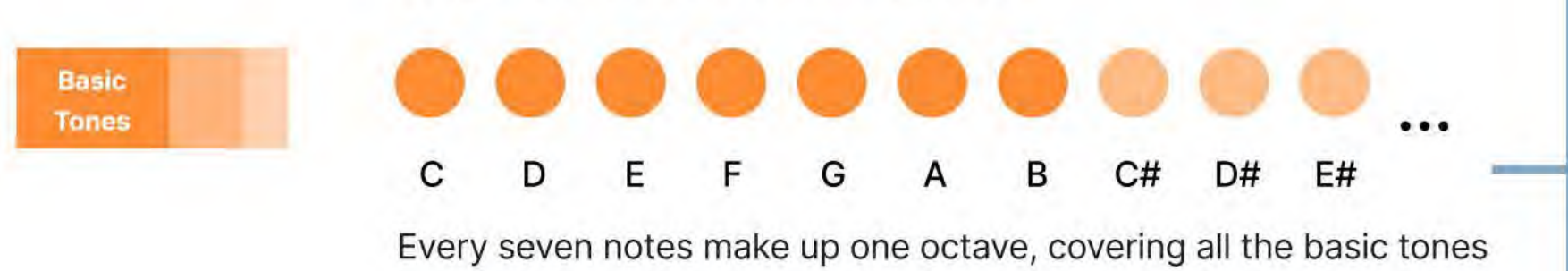
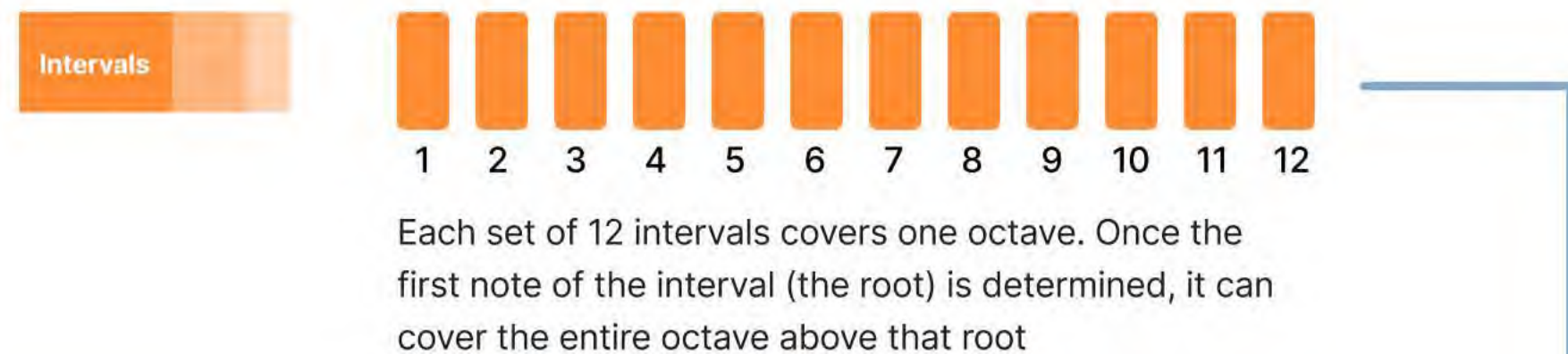
Melodic Development and Motive



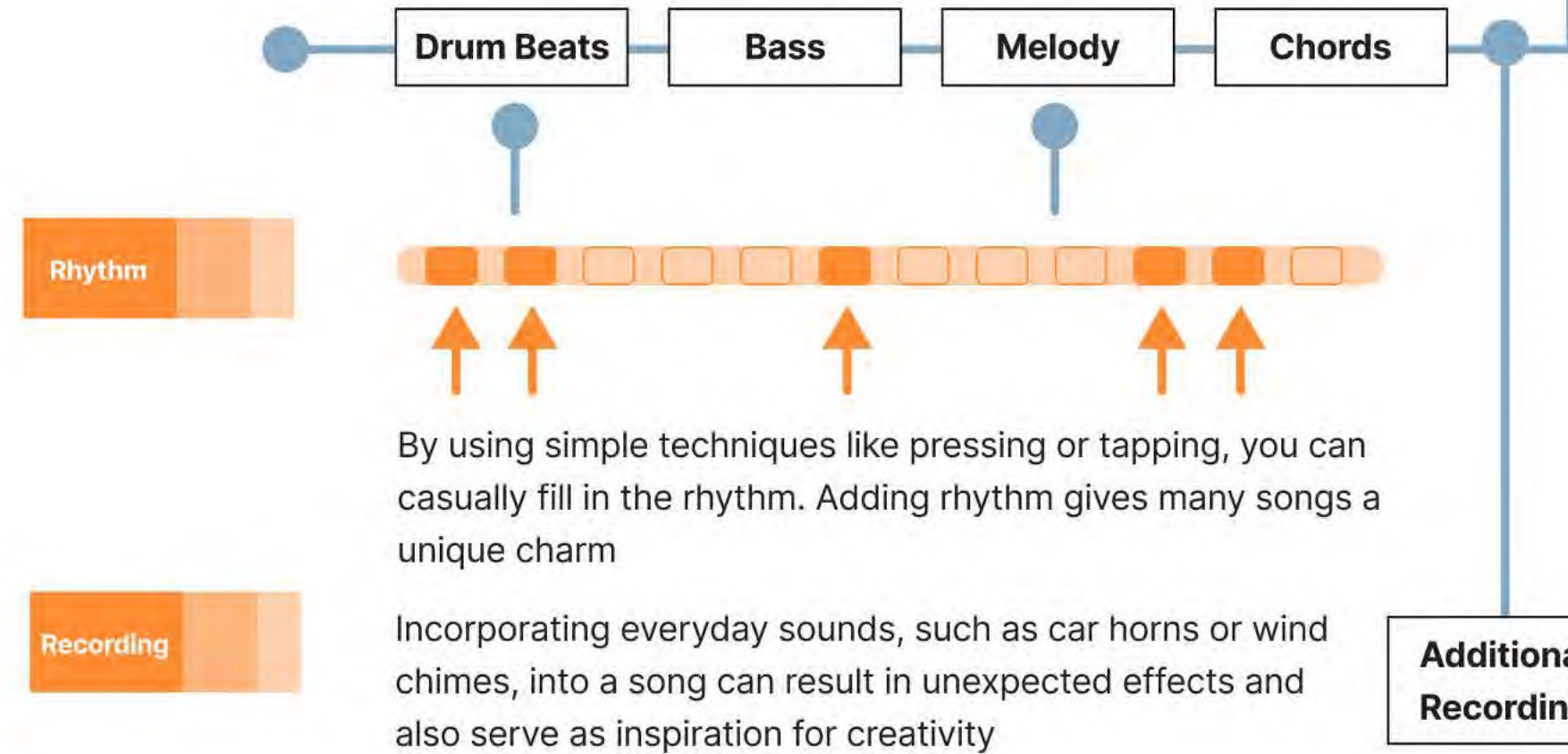
Rhythm In a piece of music, the length of each measure is fixed. Eighth notes, sixteenth notes, and so on, divide each measure into basic rhythmic beats, with each note occupying a portion of that beat



With this knowledge of music theory, you can create simple songs. Adding different instruments and ensemble arrangements can make the music more complete and rich. Many parts of composition, such as chords and accompaniment, can be seen as forms of combinations. By breaking down these musical elements, they can be transformed into individual modules, and combining different modules can produce rich results



A complete piece of music can consist of the following four parts: When intervals and basic tones are combined, they not only cover almost the entire pitch range but also allow for rich musical combinations, such as chords and arpeggios



PLATFORM INTEGRATION

1.Color Palette



2.Typeography

Aa PingFang SC	24	Banner Title
	20	Banner Title
	16	Page title, intro title, subtitle, button
	14	Body text, navigation text
	12	Body text

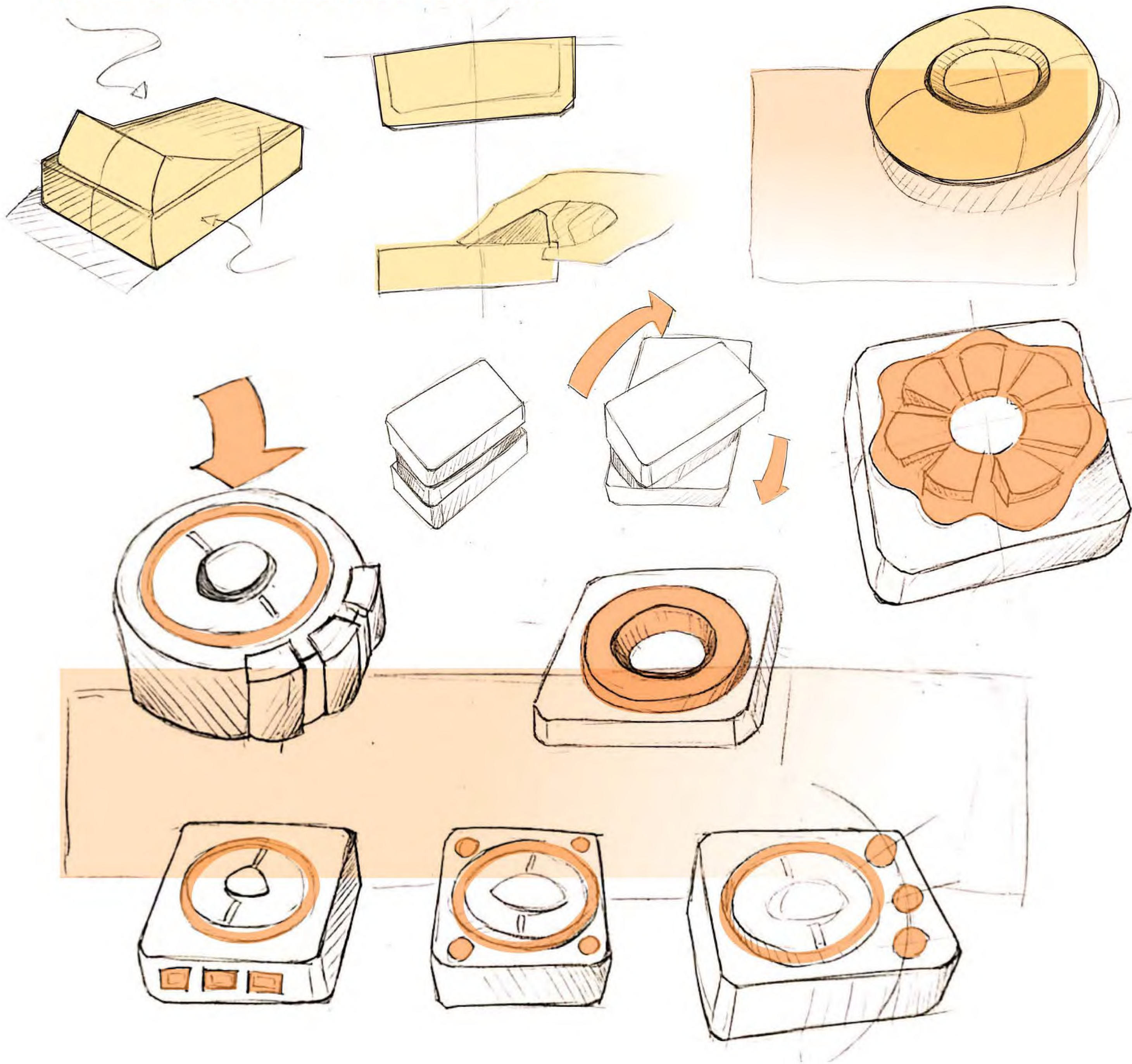
3.Color Contrast

ABCDEF	ABCDEF
ABCDEF	ABCDEF
ABCDEF	ABCDEF
ABCDEF	ABCDEF

4.Iconography

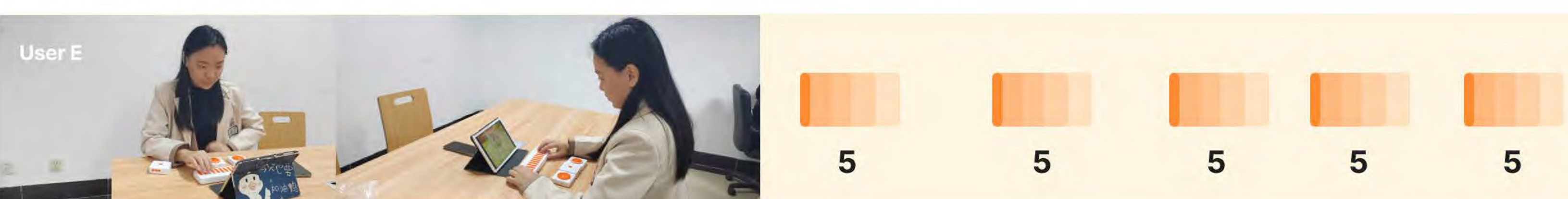
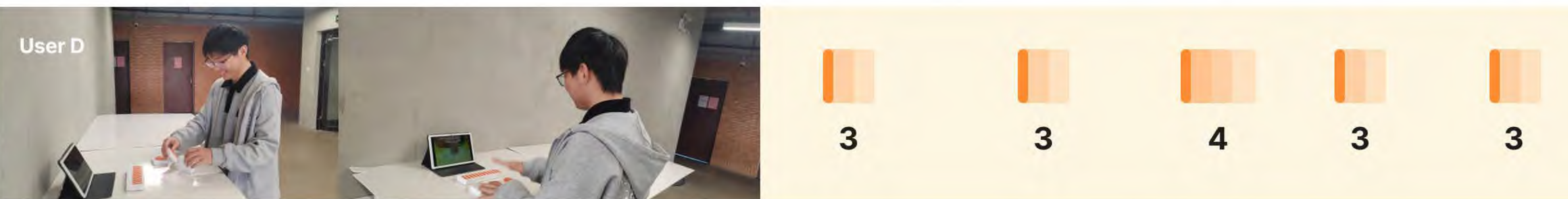
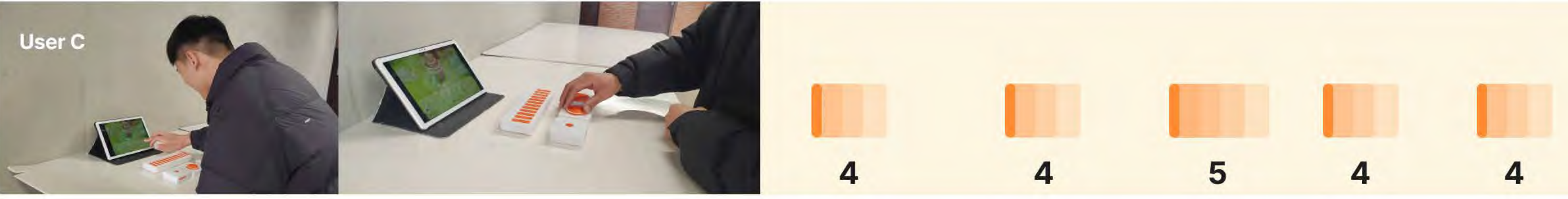
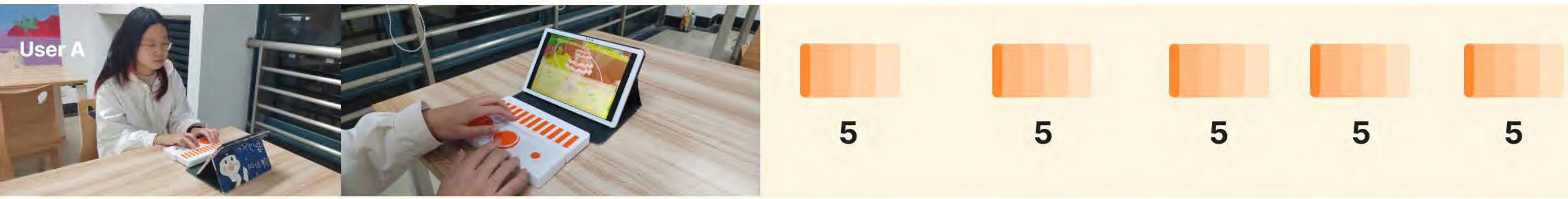


SKETCH DERIVATION



USER TEST

To study issues such as interaction complexity, page visualization, ergonomic dimensions, and creative effectiveness, I selected 5 users for testing. The testing locations included: 1. Outside a classroom on campus 2. Study room in a dormitory building 3. Activity center. After the tests, each user filled out my survey, providing valuable data.



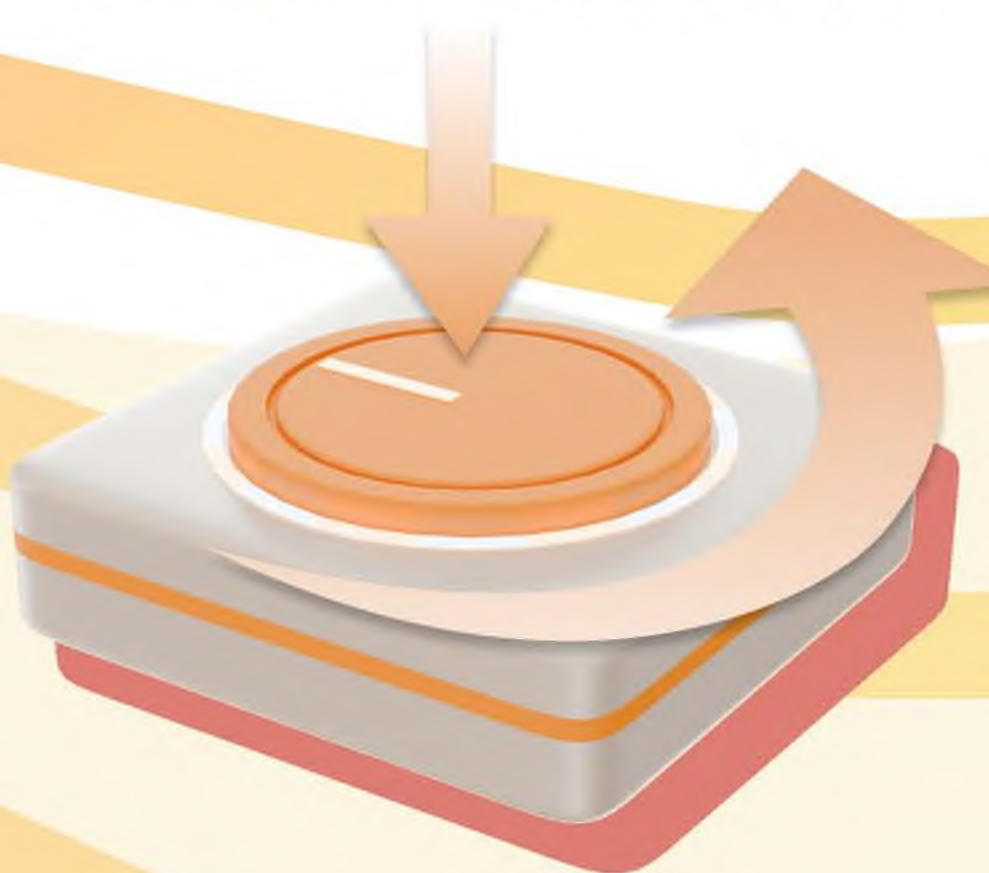
FINAL DESIGN



MODULES & UIUX

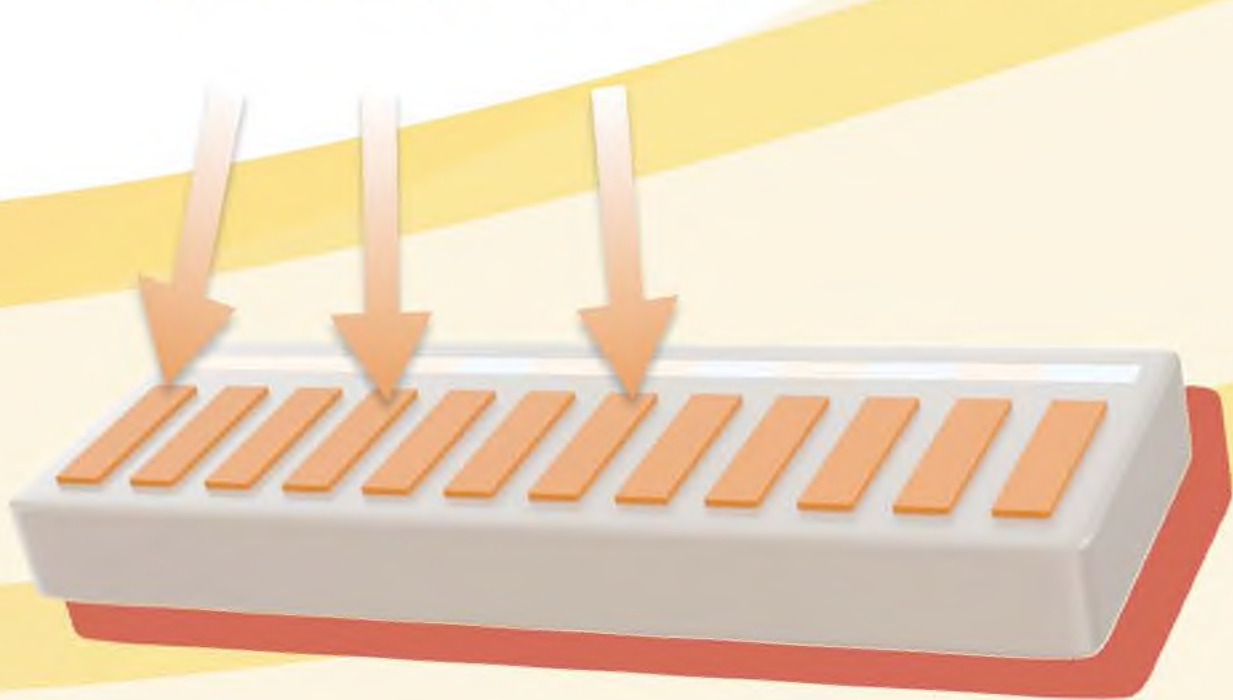
01 Basic Tone Module

This module can rotate 7 angles, each corresponding to one of the seven basic pitches (C, D, E, F, G, A, B)



02 Interval Module

12 keys correspond to 12 intervals, covering a full octave. Combined with the selection of basic tones, it can cover all pitch values



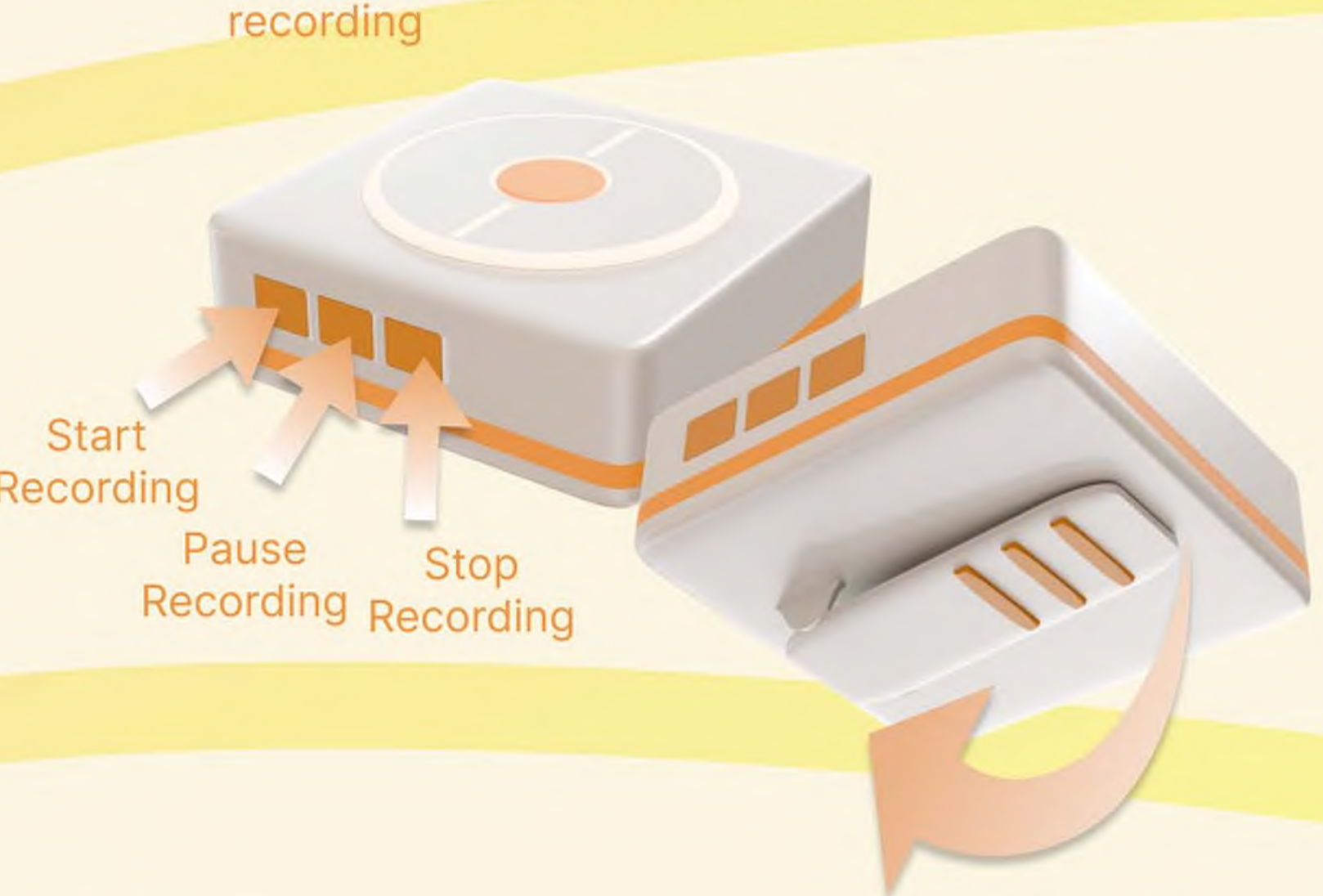
03 Rhythm Module

Press the keys in the rhythm you have in mind to record a rhythmic sequence



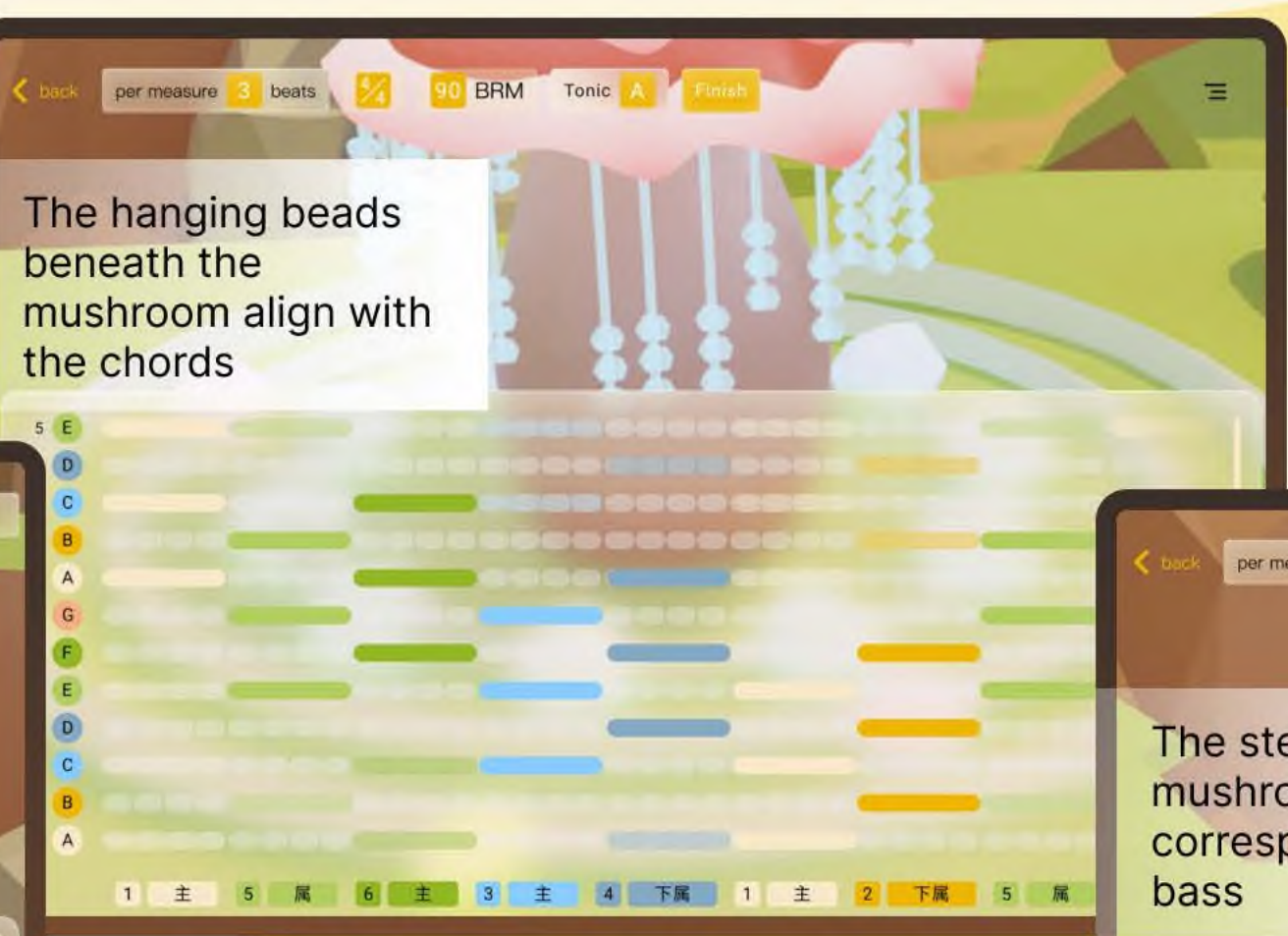
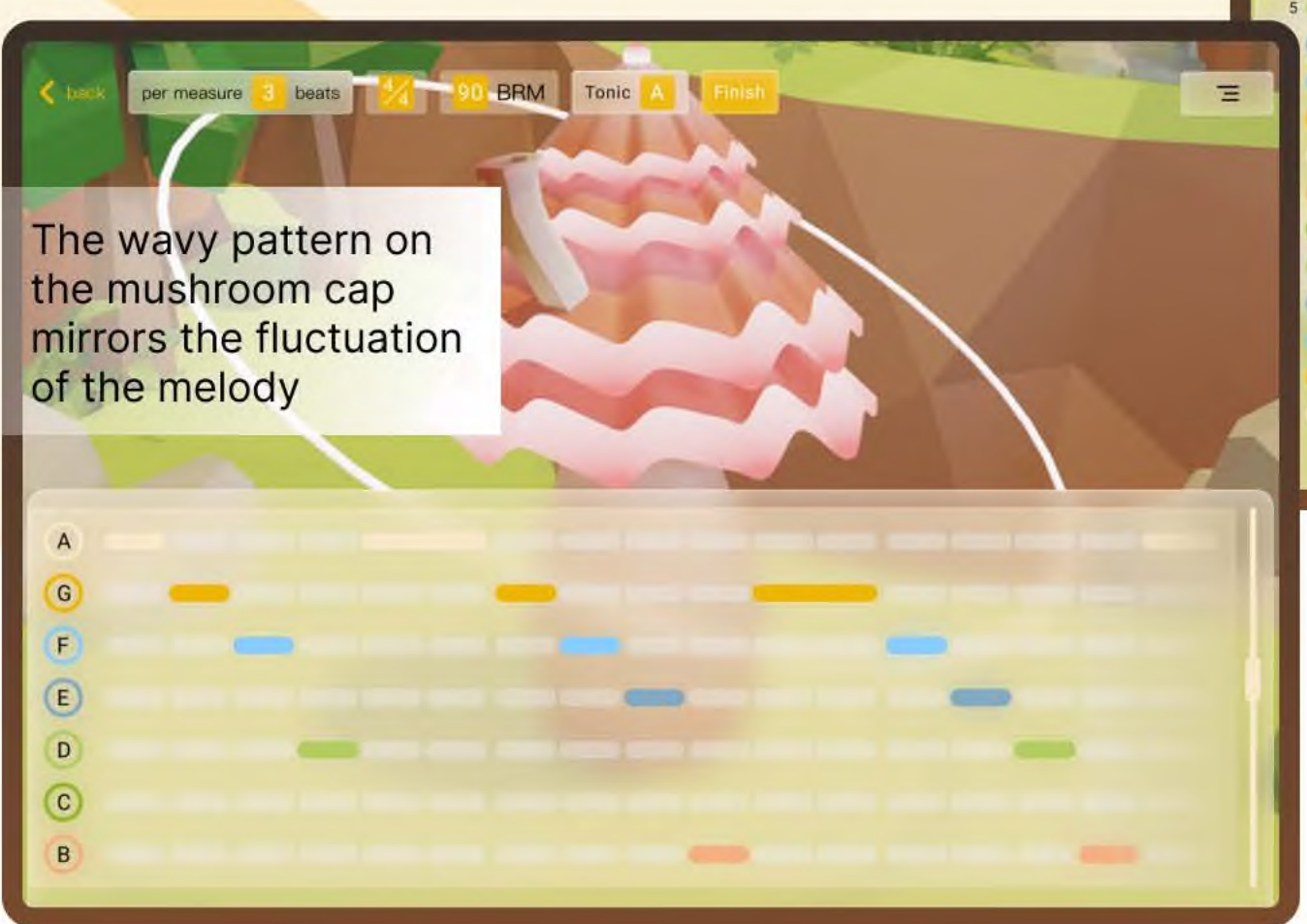
04 Recording Module

Portable for recording purposes, with a detachable clip at the bottom, allowing it to be attached to a bag or other surfaces for easy recording



Melody 01+03

By selecting basic tones and rhythms, and using functions like copying, melodies can be created

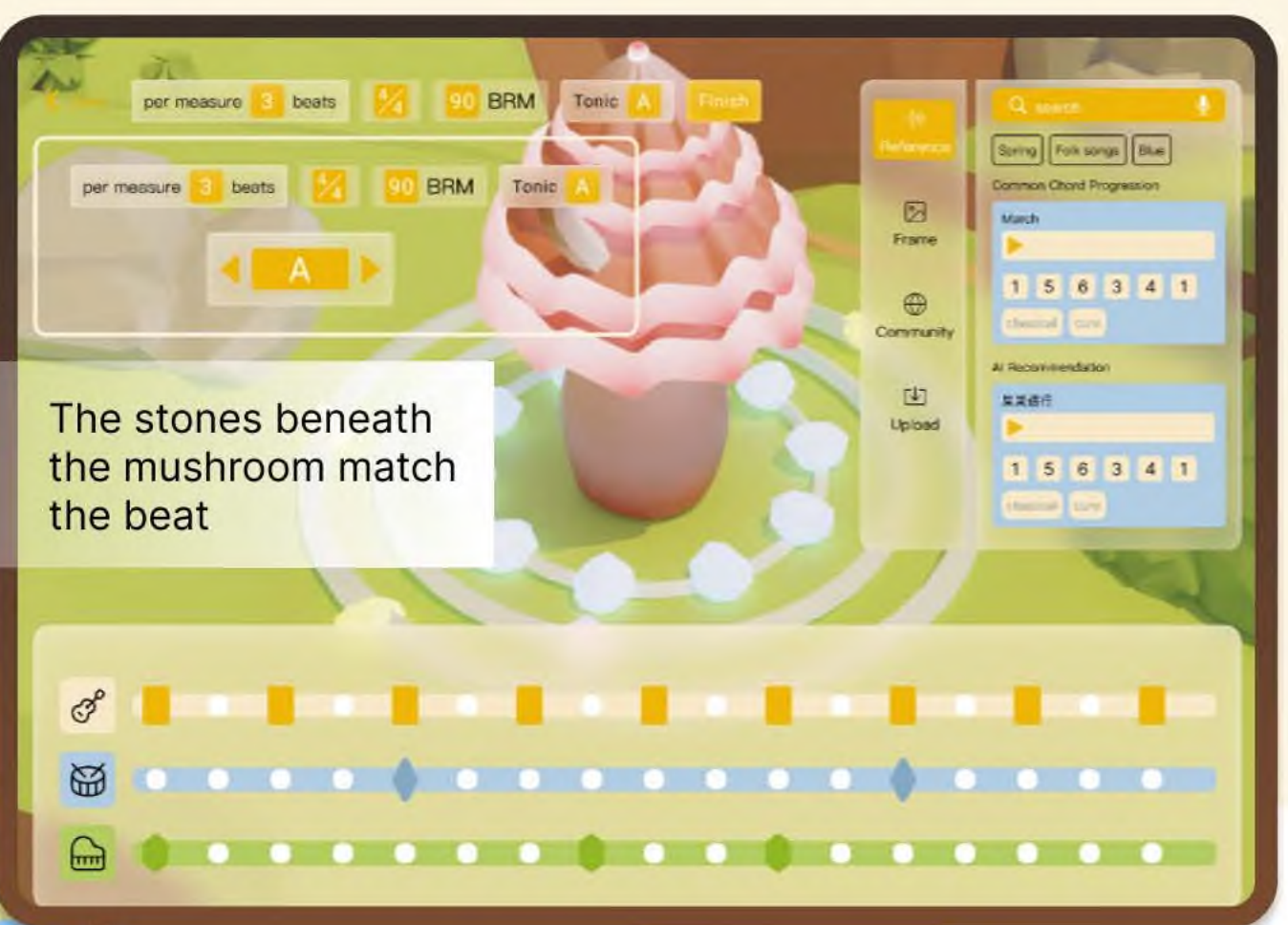


Chords 01+02

Choose the corresponding basic pitches and combine them with the interval module to compose richer combinations, such as chords

Bass 01+02

The bass can be derived by adding simple interval relationships to the basic tones of the melody



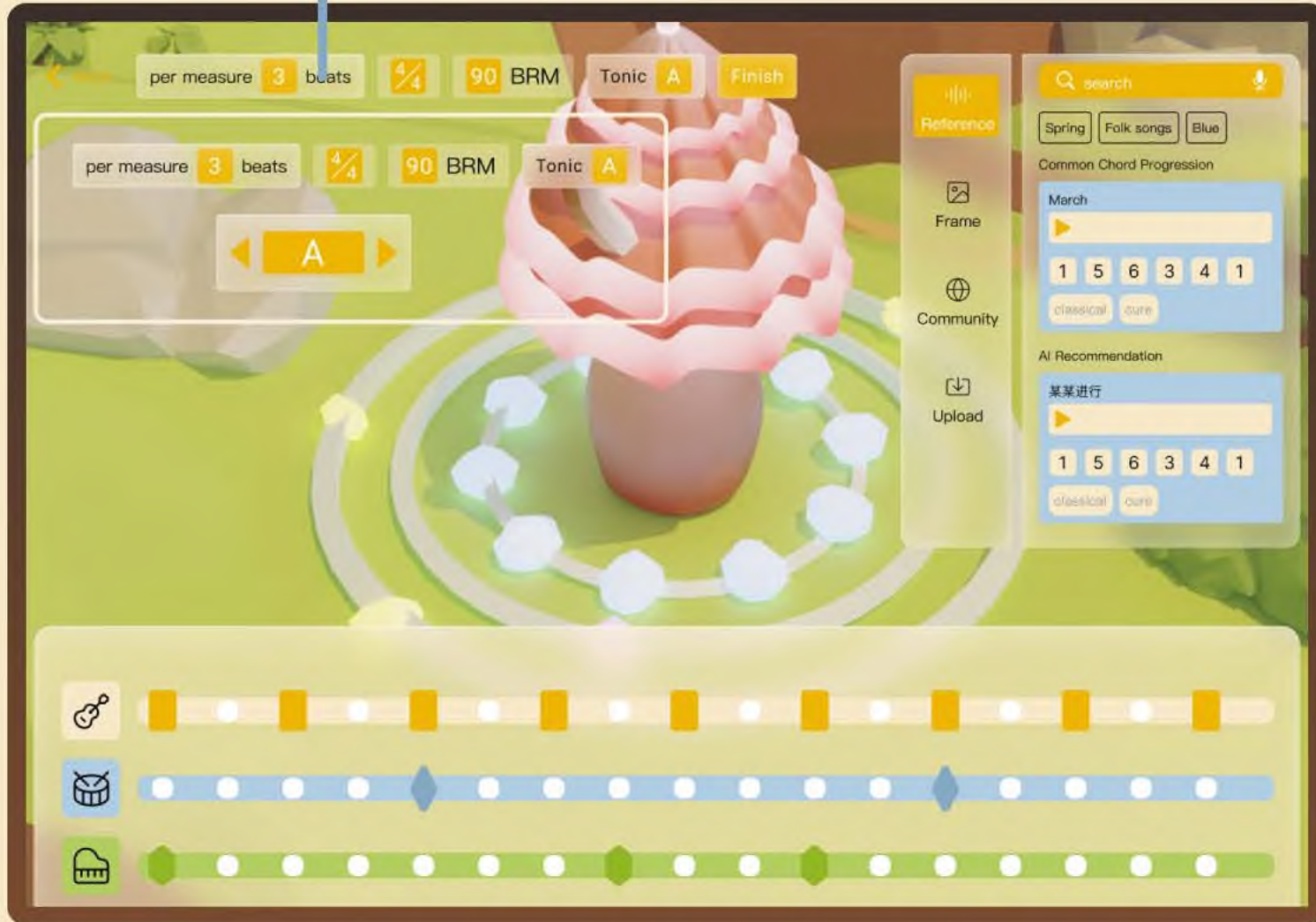
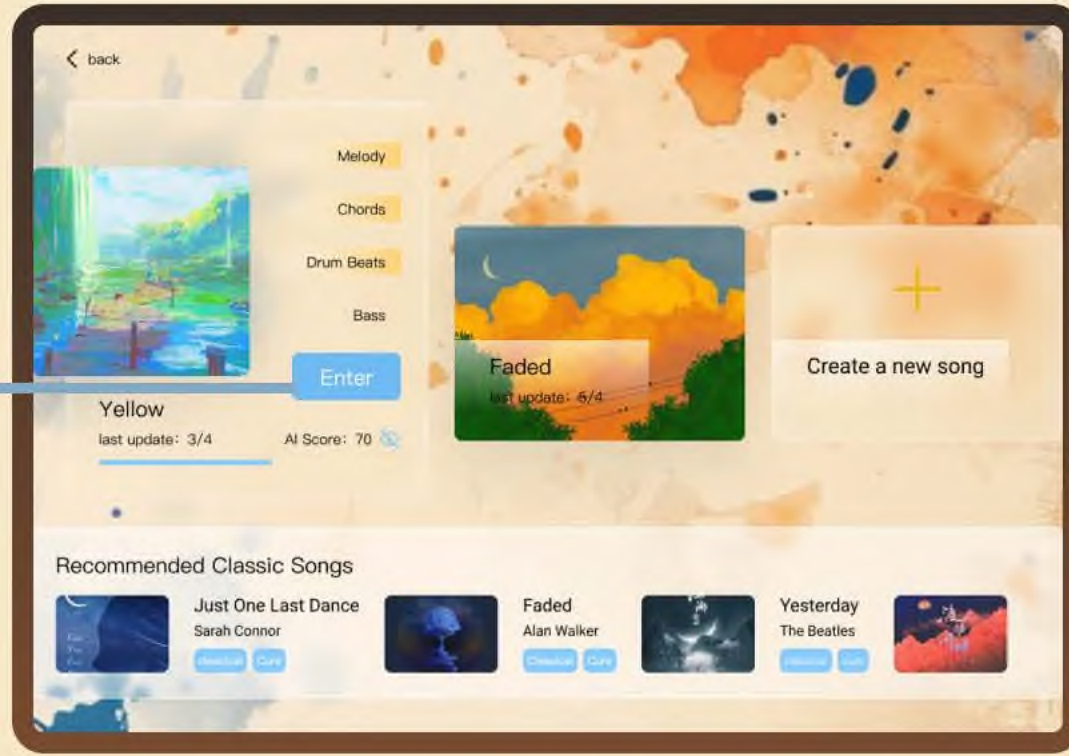
Drum Beats 03

Tap to record the corresponding rhythm, which can be used as drum beats

Song Selection Page

Create a new song or refine an existing one here.

Adjust basic settings such as the song's main melody and tempo.

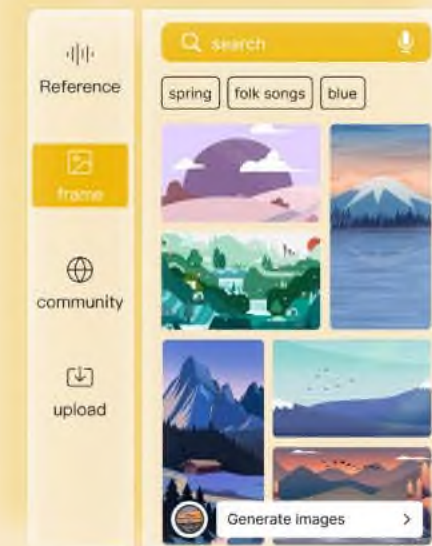


Find song references and inspiration through publicly shared songs or AI recommendations.



Reference

Search for desired musical styles and receive recommended classic melodies as composition references



Frame

Choose visual themes based on desired composition style, and have AI generate melodies to assist in creation



Community

View songs publicly shared by others in the community for reference and use

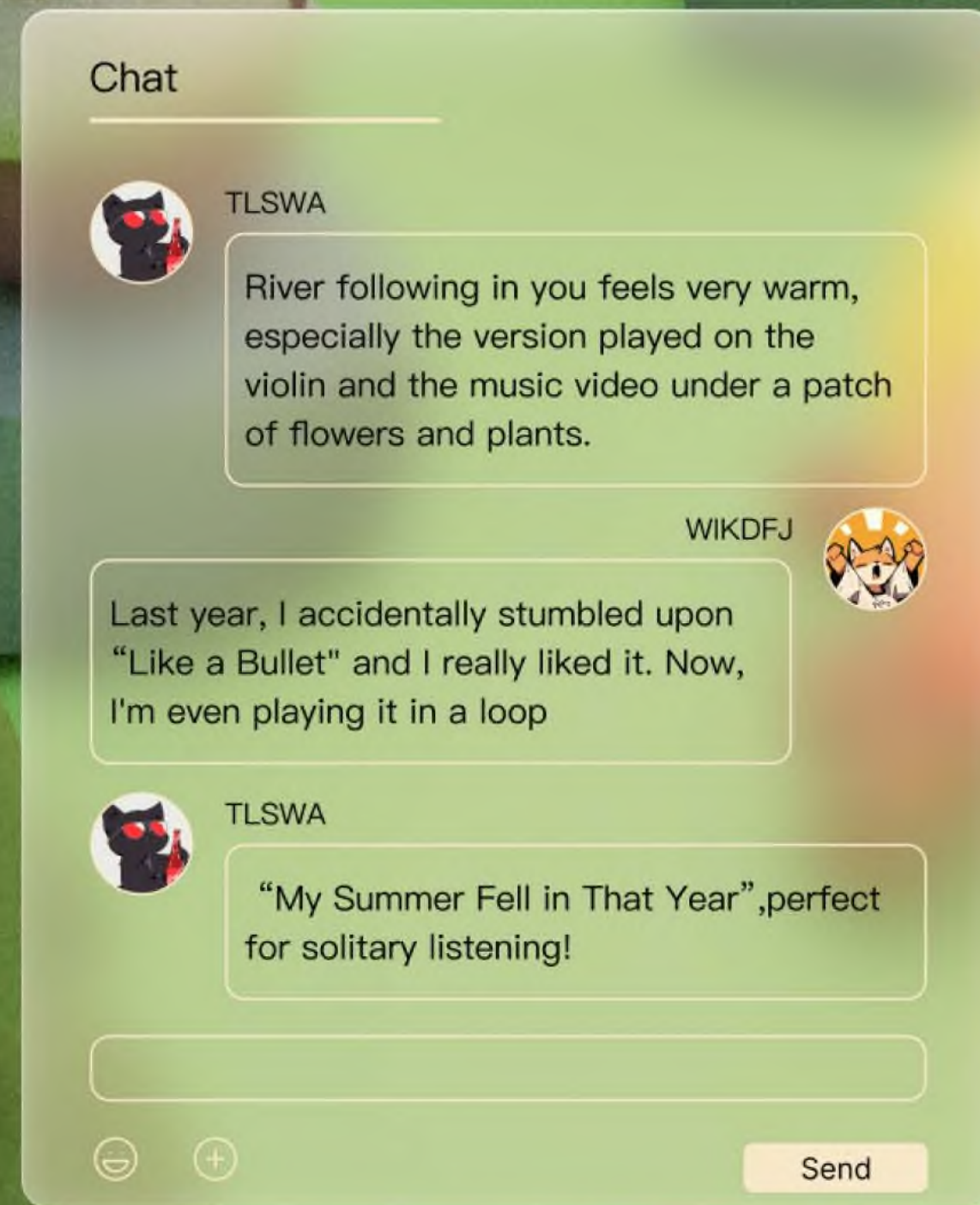


Upload

Download and add songs for personal use

Multi-User Public Space

Music Salon



Function 1 Music Salon

In public spaces, users can initiate or join a music salon. Participants can gather around a virtual campfire to share music-related content, including favorite songs, creative inspiration, and stories.

Function 2 Background Music

Users can publish their original songs in public spaces, where the songs appear as models. Other users can play them as background music.

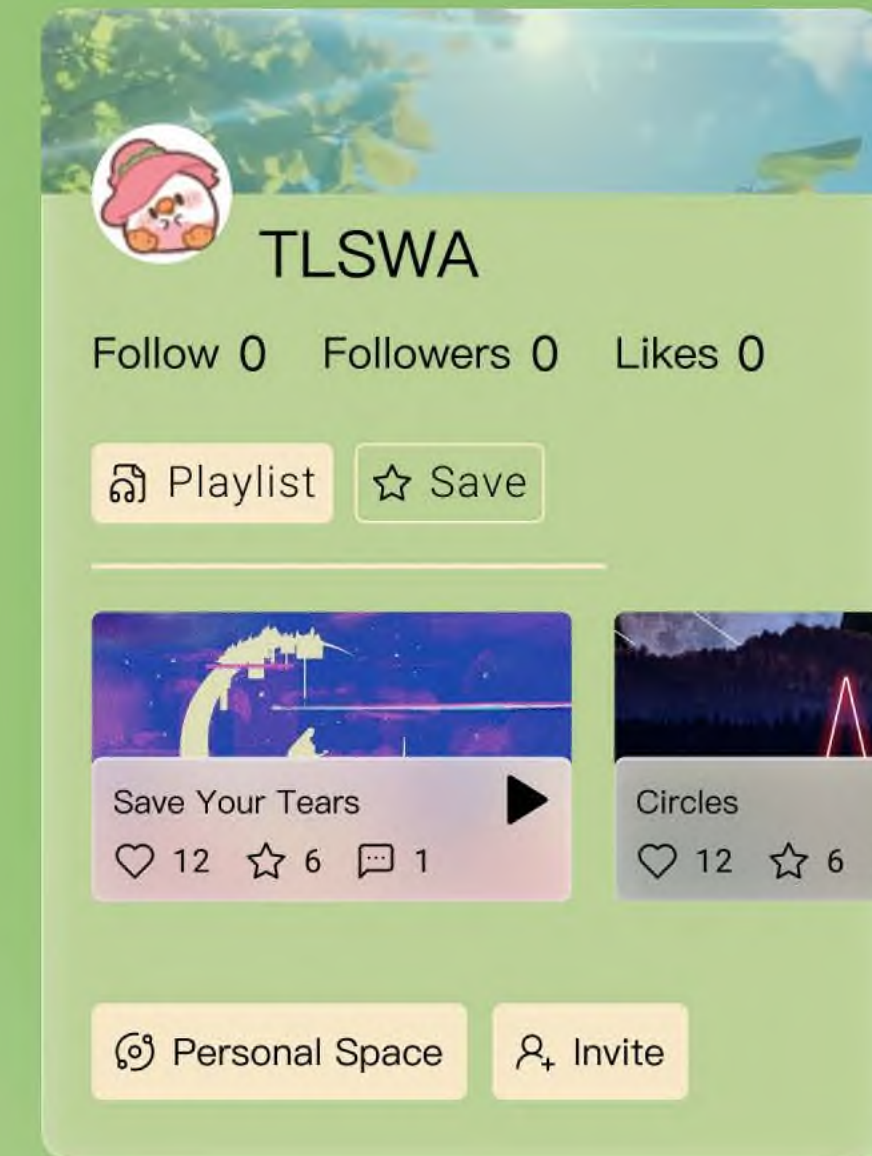
Function 4 Collaborative Composition

Users can start or join a collaborative composition for a song, contributing to the creation of different parts of the track.

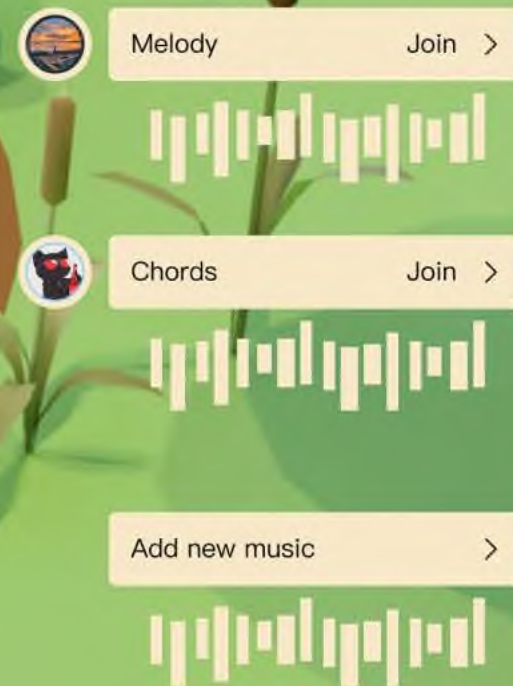
Function 1

View and Add

During interactions in the public space, users can view each other's profiles, see publicly shared playlists, and send friend requests.



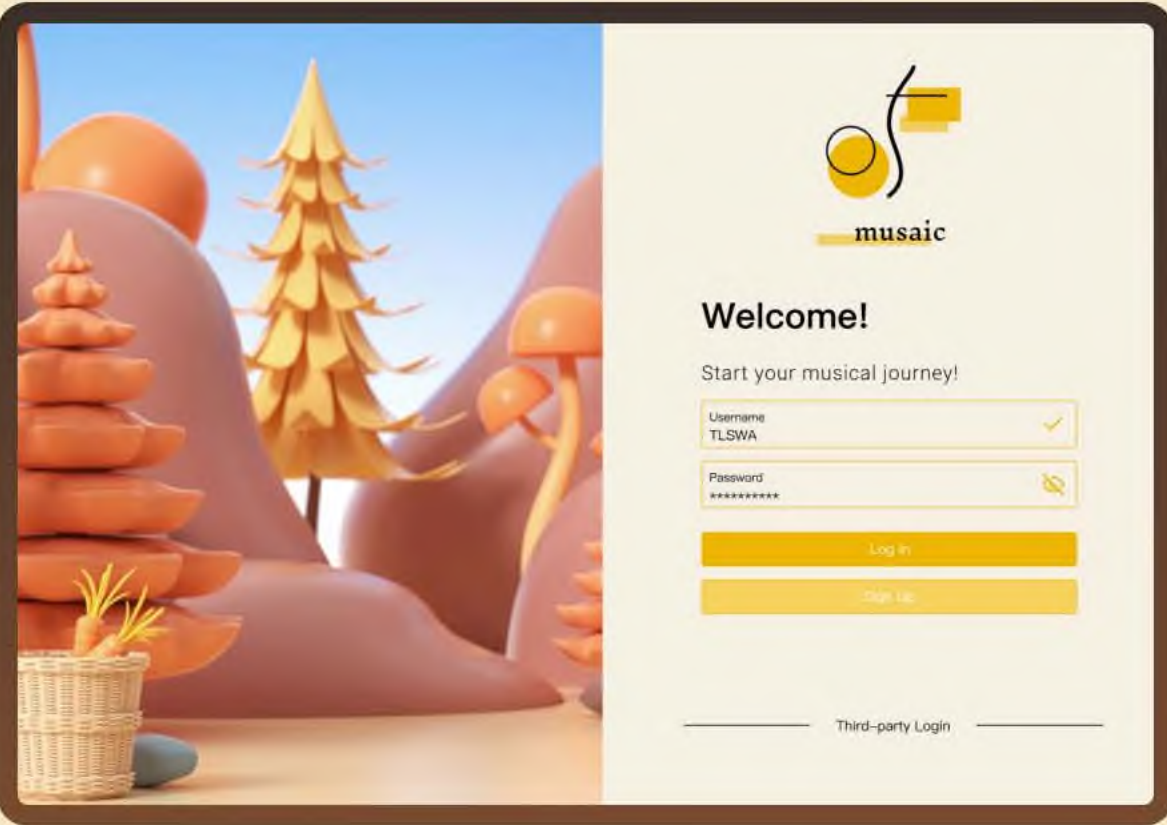
Collaborative arrangement online



Play

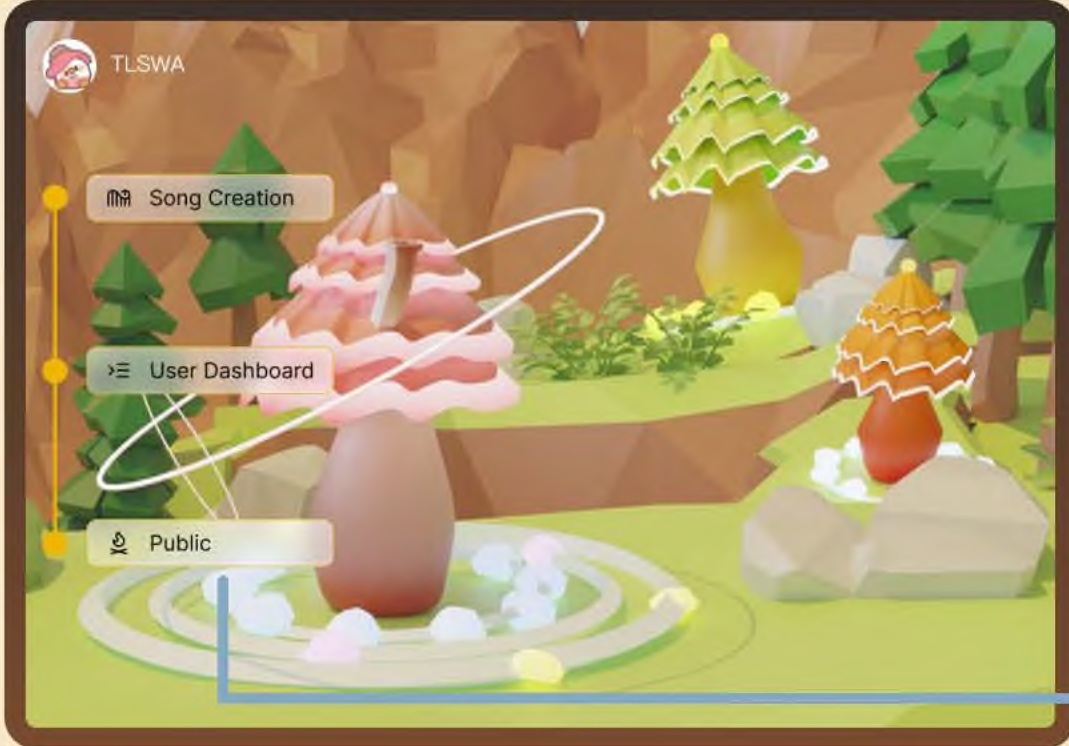


UX FRAME



Login/Register Page

Register or log in to access the personal homepage.



Song Creation

User Dashboard

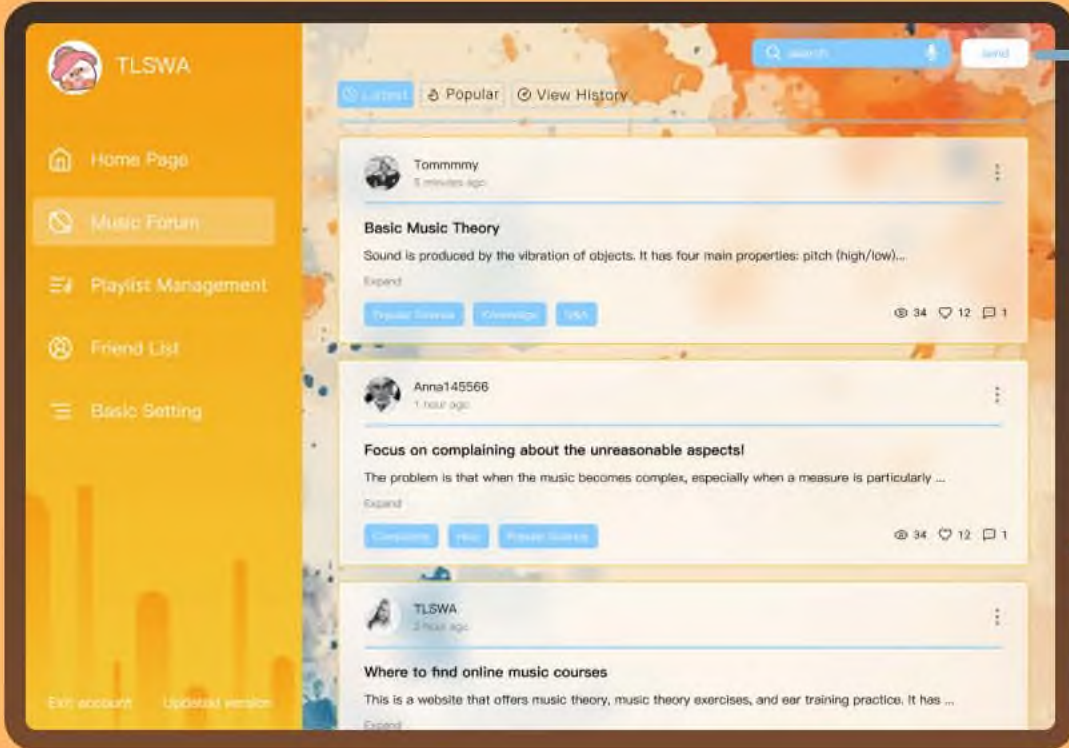
Personal Homepage

The personal homepage displays all created songs, presented as visualized plants within the space. From here, you can also access the other three functional areas.

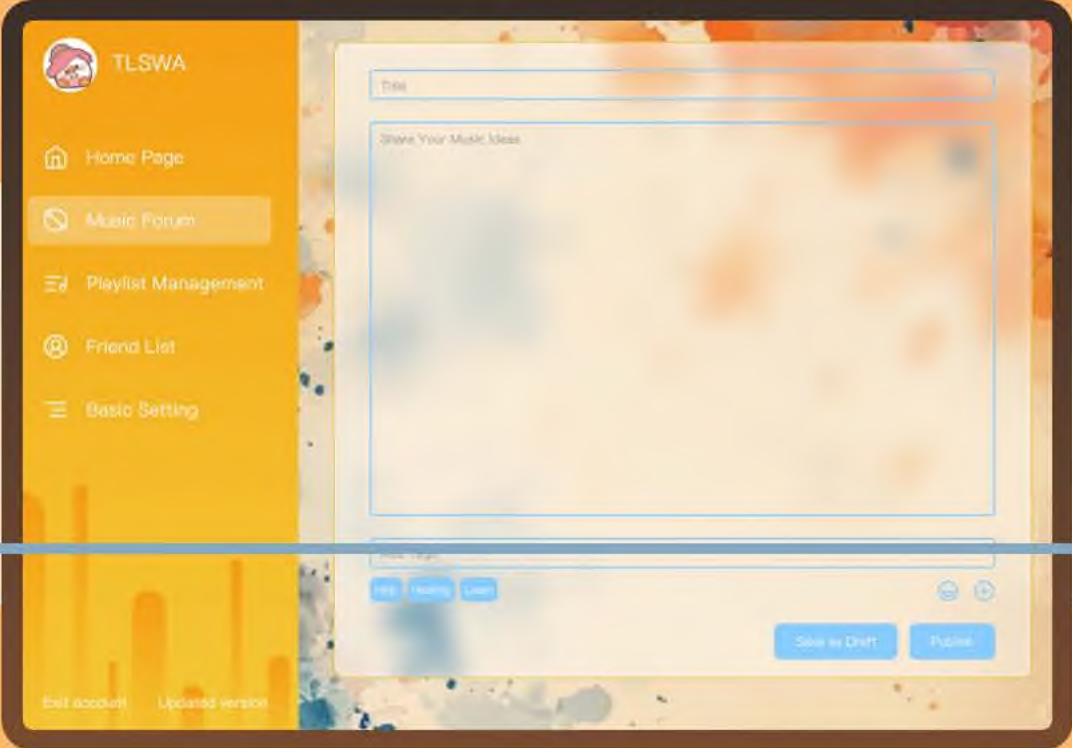
Public Space

1 Music Forum

Within the forum, you can view various types of posted information, including educational content, requests for help, etc.

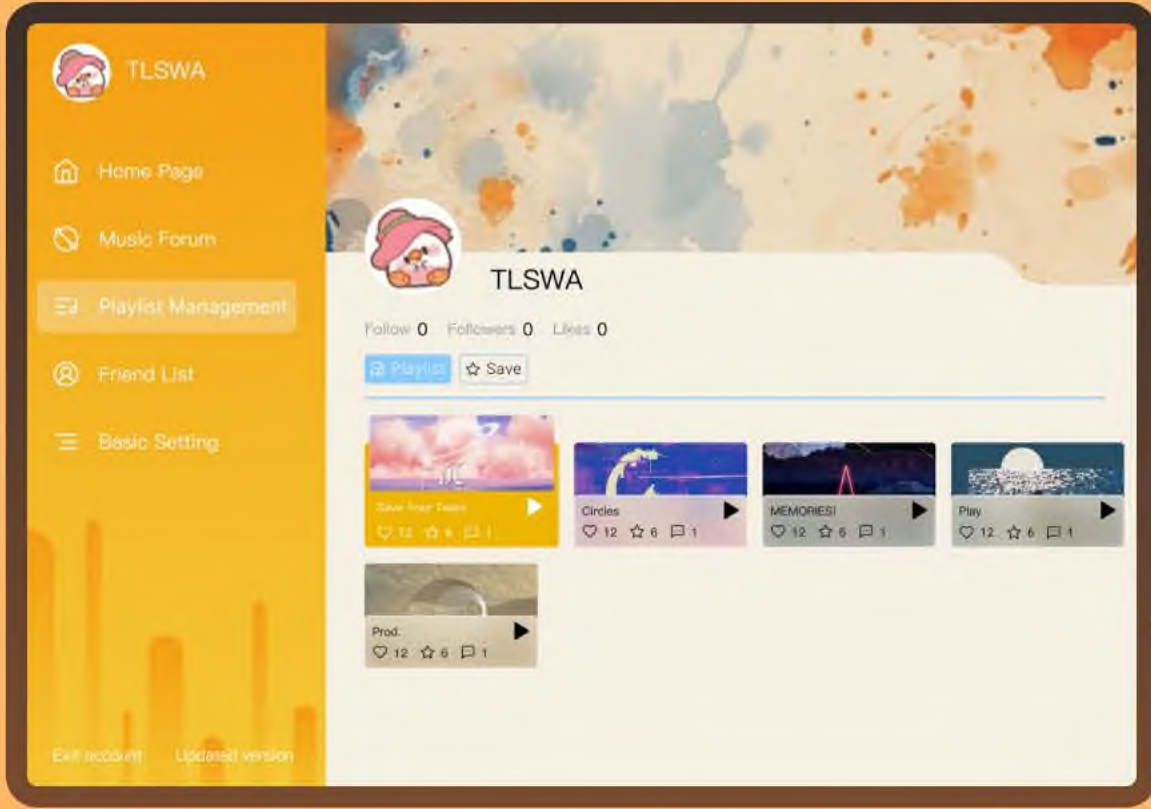


Click on a post to view its full content and all replies within the post.

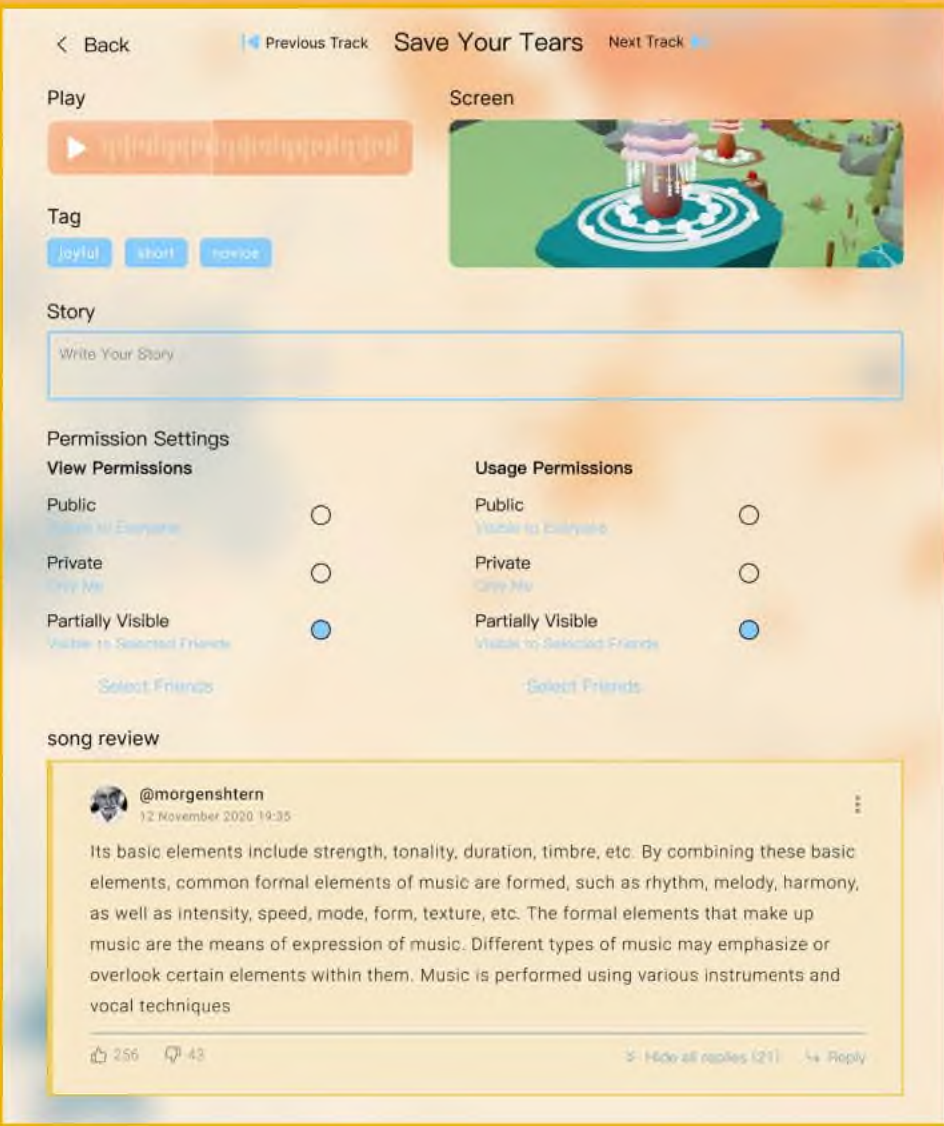
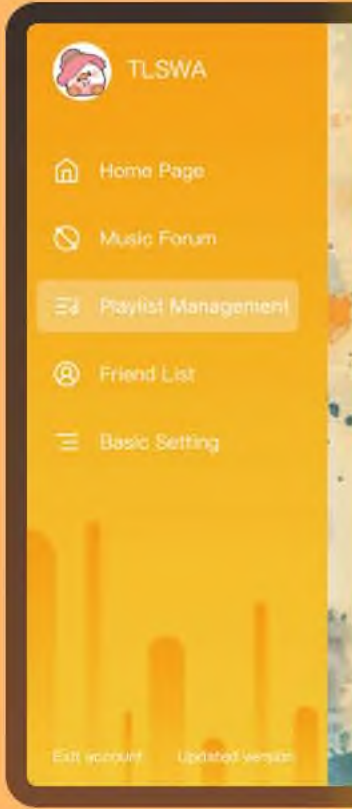


Publish your own posts, write the title, content, and add tags for easier categorization.

2 Playlist Management



View and manage your personal song library, and adjust settings.

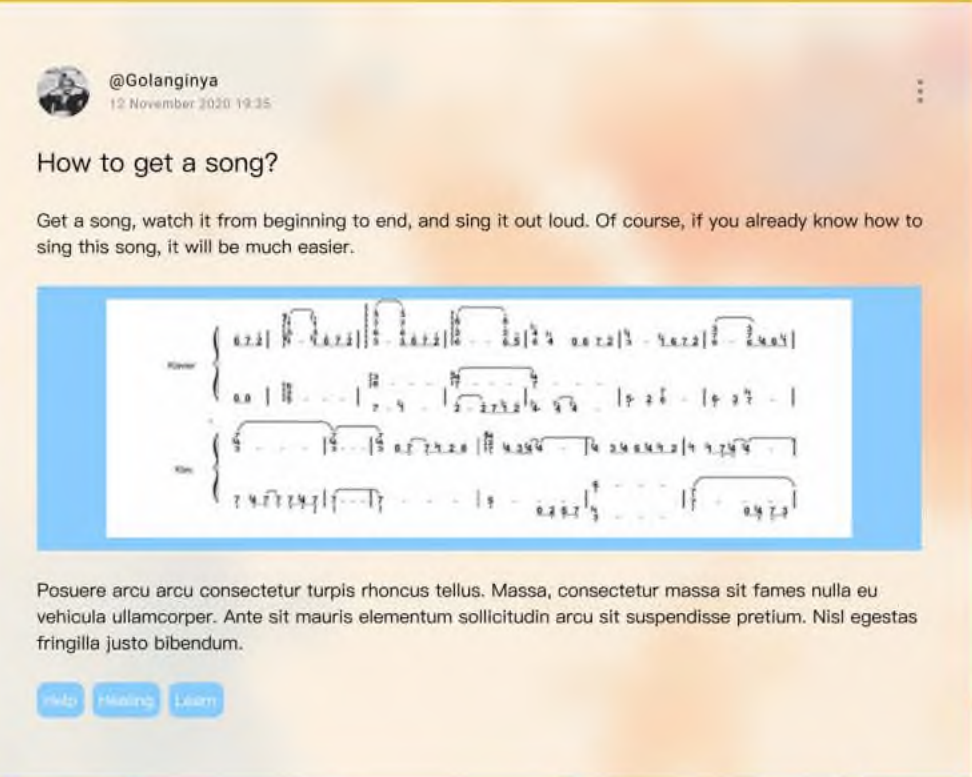


Song Information Settings

View the music, model, and write the background story for each individual song.

Song Permissions Settings

View Song Comments



View Full Post Content

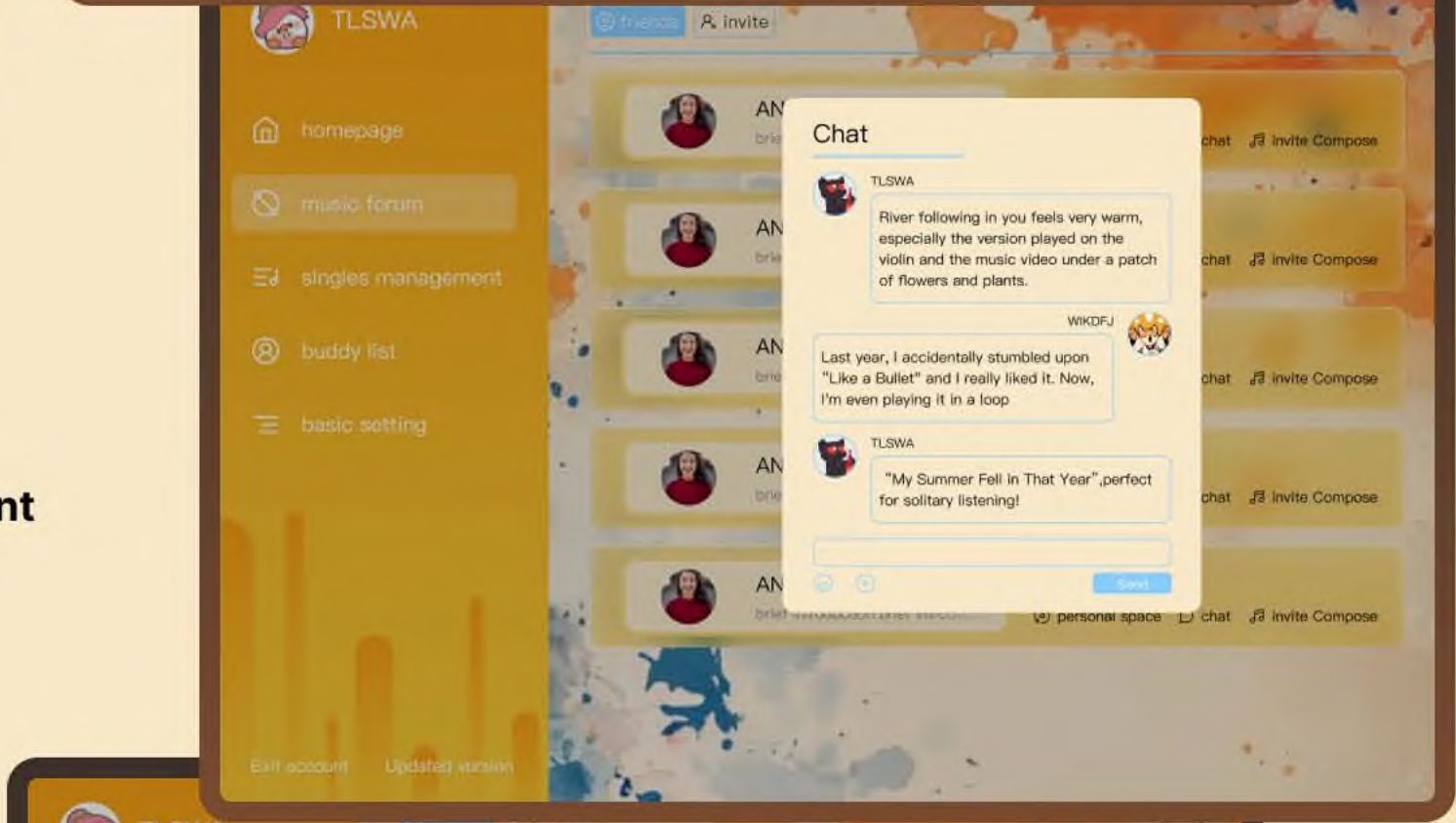
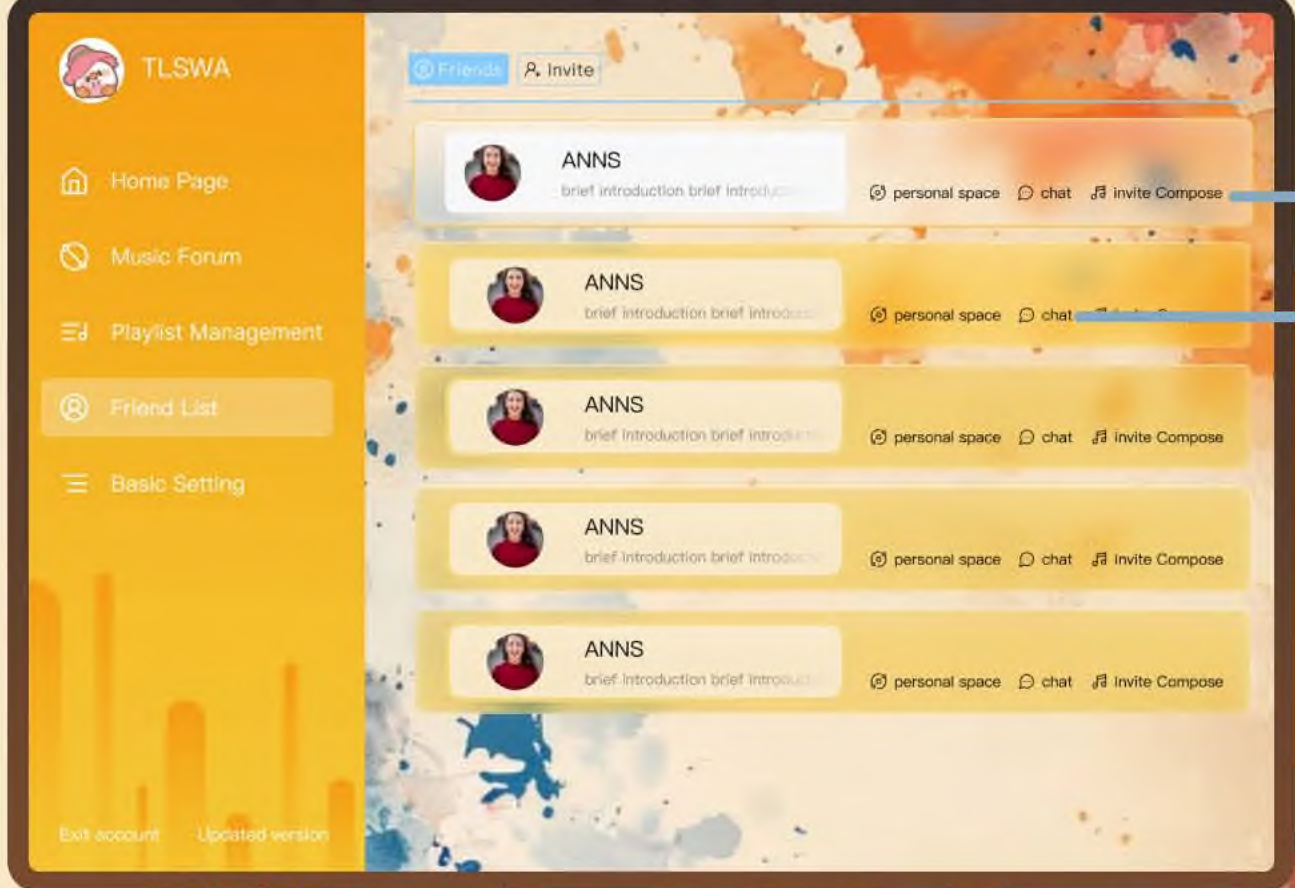


Reply to Post

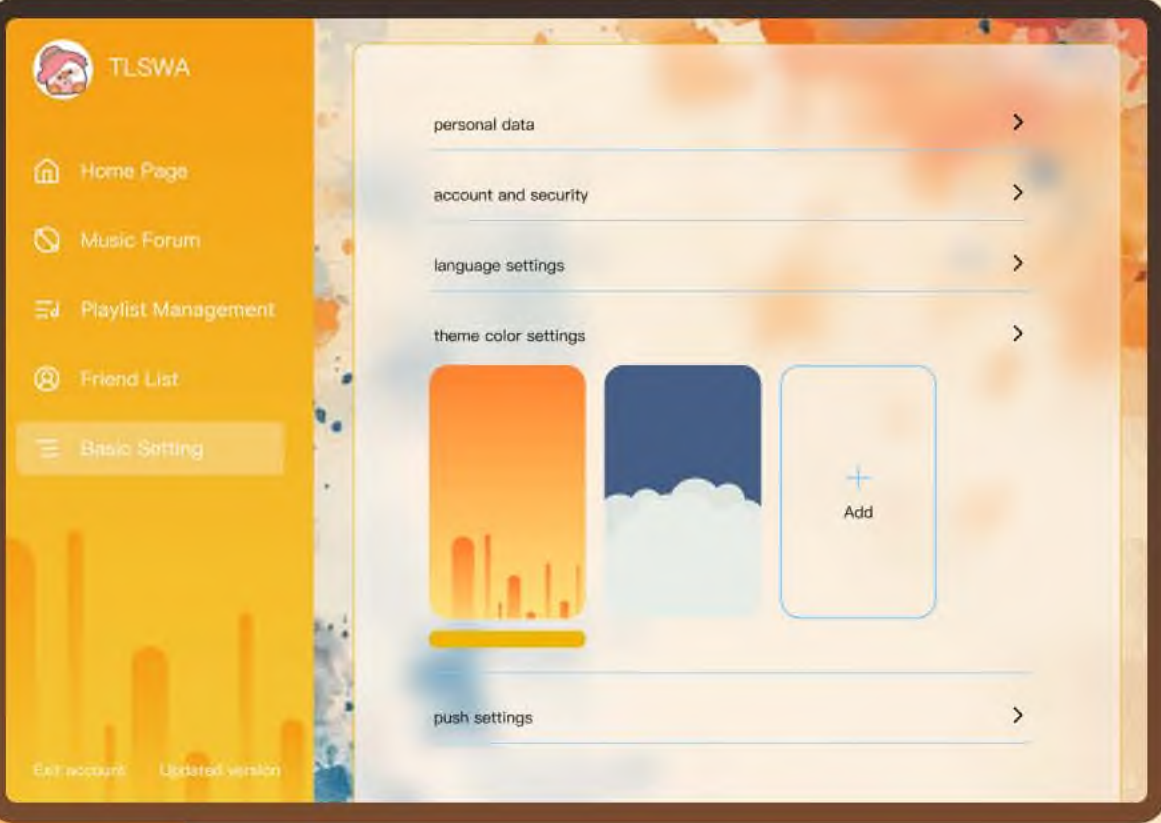
View Other Replies

3 Friend List

Here you can view, manage, and use the friend list to chat with friends and access related features.



4 Basic Setting



In the settings, you can modify various app settings, including changing the app's color scheme and background, for a more personalized experience.

This page allows quick access to the chat interface and view chat history.

Invite friends to compose music by sending existing songs for continued creation or co-create new songs together.